

Cuánticos para Clasificación de Imágenes

Estancia Sebastián BCAM // Junio - Julio

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- Experimento 1.

Análisis del espacio Latente del Bottleneck de la UNET y de CNNs

- Experimento 2.

- Experimento 3.

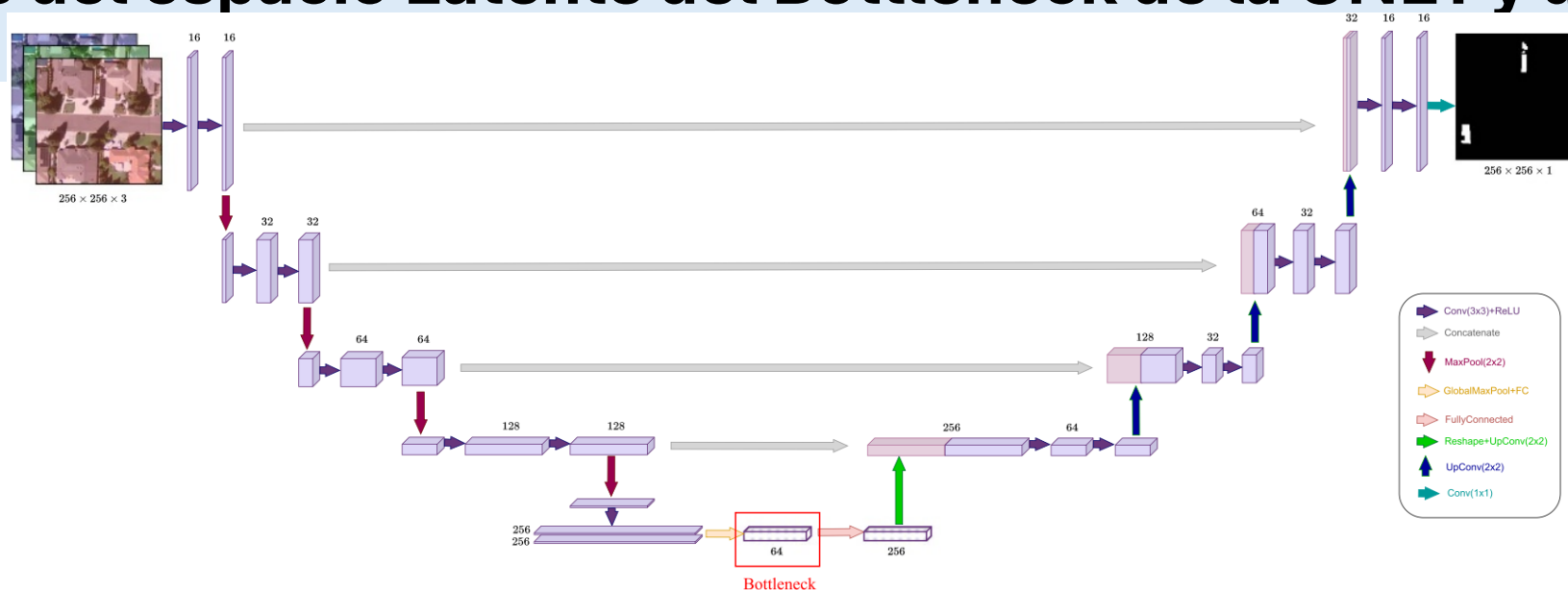
- Experimento 4.

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Análisis del espacio Latente del Bottleneck de la UNET y de CNNs



P. Rodriguez-Grasa, R. Farzan-Rodriguez, G. Novelli, Y. Ban, and M. Sanz, "Satellite image classification with neural quantum kernels," *Machine Learning: Science and Technology*, vol. 6, no. 1, Art. no. 015043, Feb. 2025, doi: 10.1088/2632-2153/ada86c.

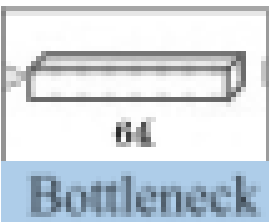
Técnicas de reducción de dimensionalidad sobre Z_{64}

PCA	ICA
AE-Lineal	AE-No Lineal

Aprendizaje de Representaciones Latentes en imágenes



Técnicas de reducción de dimensionalidad sobre Z_{64}



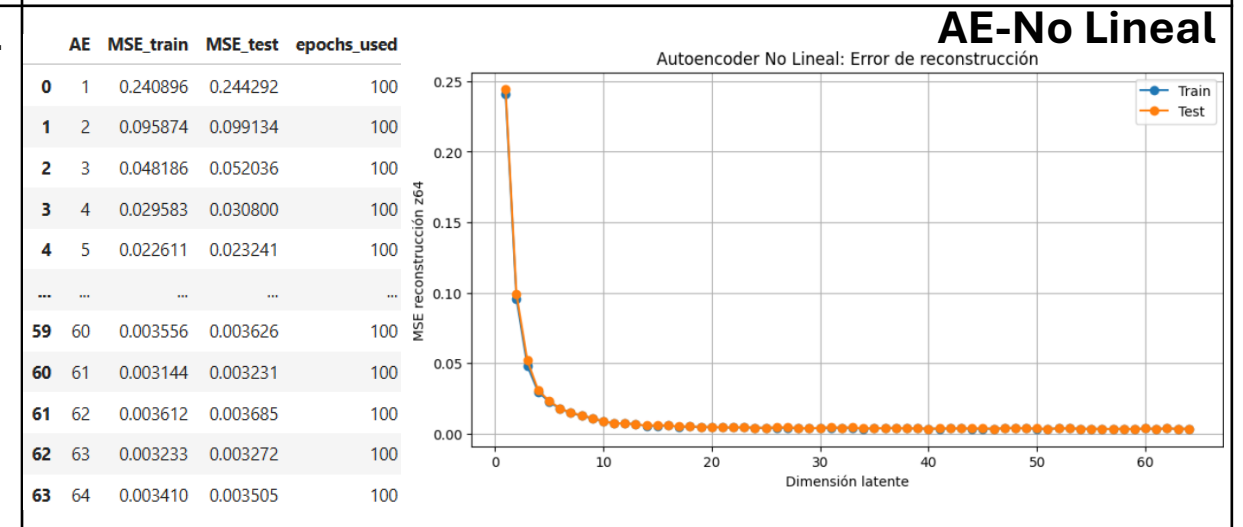
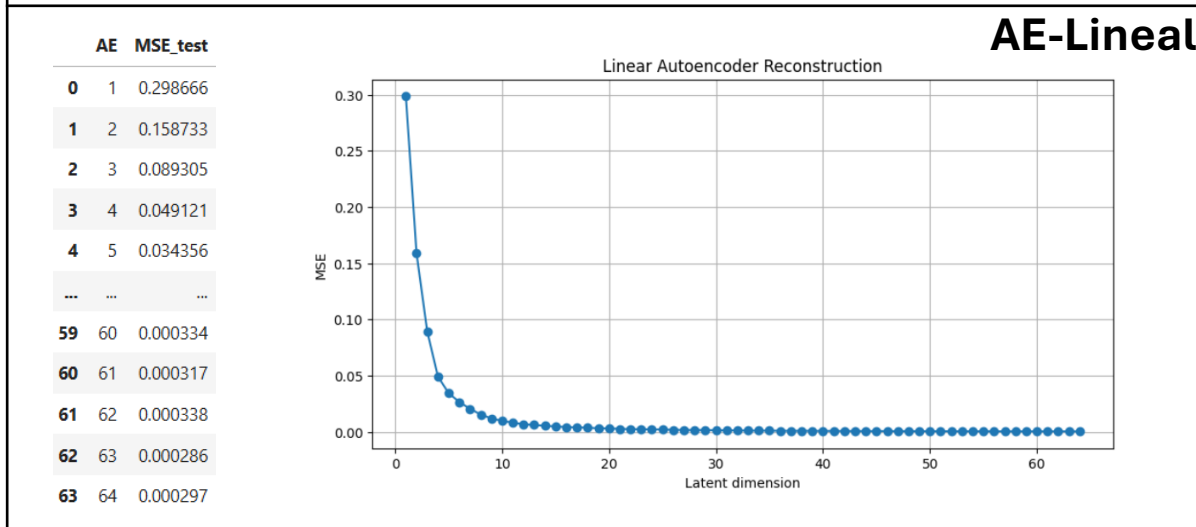
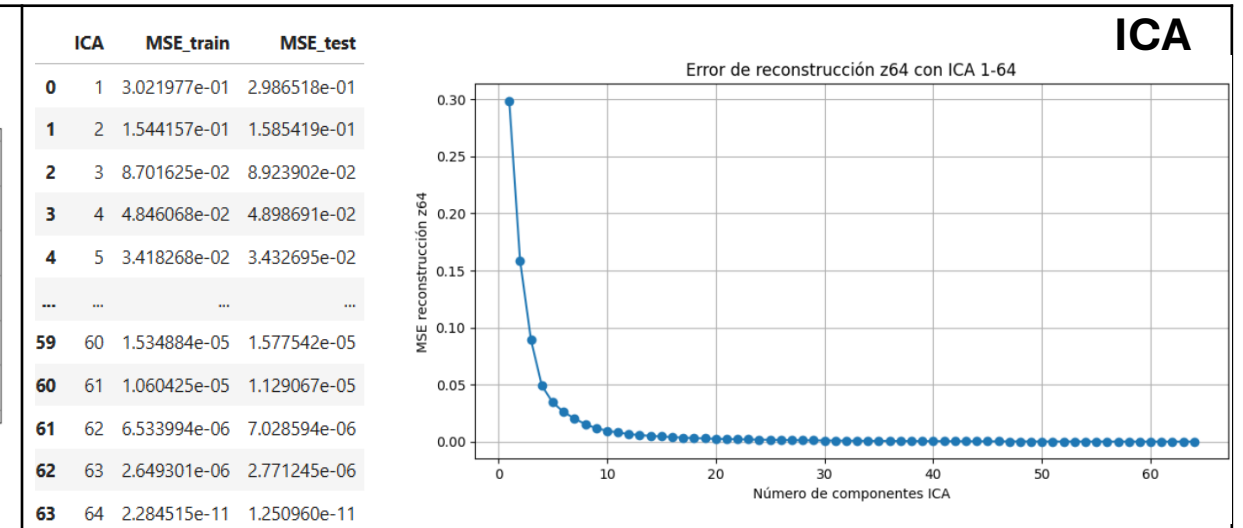
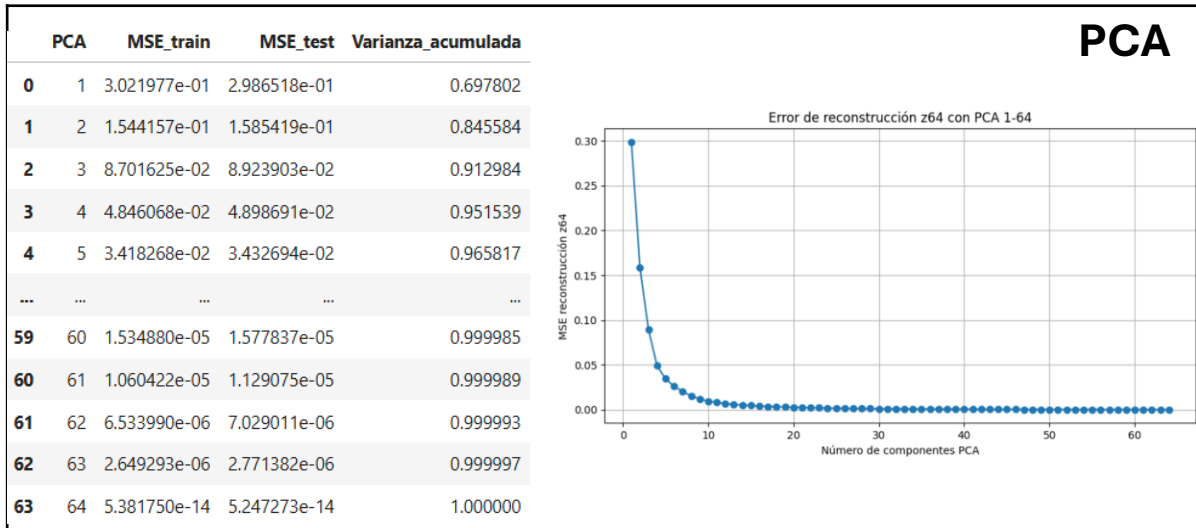
Método	¿Cómo reduce de 64 a 3 características?	¿Qué aprende durante el entrenamiento?	Salida utilizada
PCA	Calcula una nueva base de coordenadas y proyecta cada muestra sobre las 3 direcciones con mayor varianza .	Las 3 direcciones principales que explican la mayor variabilidad del conjunto de entrenamiento.	3 componentes principales.
ICA	Calcula una nueva base donde las variables sean lo más independientes posible y proyecta cada muestra sobre esas tres direcciones.	Las 3 componentes independientes aprendidas del conjunto de entrenamiento.	3 componentes independientes.
AE-Lineal	Una red neuronal reduce progresivamente las 64 variables a 3 neuronas mediante una única transformación lineal (64→3). Después intenta reconstruir las 64 variables (3→64) para comprobar cuánto se perdió durante la compresión.	Los pesos del Encoder (64→3) que permiten conservar la mayor cantidad posible de información minimizando el error de reconstrucción (MSE).	Solo las 3 neuronas del Encoder. El Decoder se elimina una vez finalizado el entrenamiento.
AE-No Lineal	Una red neuronal primero transforma las 64 variables en 32 neuronas ocultas , aplica una función de activación ReLU , y después las comprime hasta 3 neuronas (64→32→3) . Posteriormente reconstruye las 64 variables (3→32→64) para medir la pérdida de información.	Los pesos del Encoder (64→32→3) , capaz de aprender relaciones no lineales entre las variables gracias a la capa oculta y a la activación ReLU .	Solo las 3 neuronas del Encoder. El Decoder se elimina una vez finalizado el entrenamiento.



Etapa	PCA	ICA	AE-Lineal	AE-No Lineal
Entradas de entrenamiento	TRAIN (64D)	TRAIN (64D)	TRAIN (64D)	TRAIN (64D)
Normalización	StandardScaler.fit() → transform(TRAIN)			
Aprendizaje (solo TRAIN)	PCA.fit(TRAIN_std)	FastICA.fit(TRAIN_std)	Entrenar Encoder (64→3) + Decoder (3→64) minimizando MSE(TRAIN_std, Reconstrucción)	Entrenar Encoder (64→32→3) + Decoder (3→32→64) minimizando MSE(TRAIN_std, Reconstrucción)
Extracción de características	PCA.transform(TRAIN_std) → Z_train (3F)	FastICA.transform(TRAIN_std) → Z_train (3F)	Encoder(TRAIN_std) → Z_train (3F)	Encoder(TRAIN_std) → Z_train (3F)
Escalado angular	MinMaxScaler.fit(Z_train) → transform(Z_train)			
¿Cuándo entra TEST?	Aquí comienza el conjunto de prueba	Aquí comienza el conjunto de prueba	Aquí comienza el conjunto de prueba	Aquí comienza el conjunto de prueba
Normalización de TEST	transform(TEST) usando μ y σ aprendidas de TRAIN			
Obtención de las 3 características de TEST	PCA.transform(TEST_std)	FastICA.transform(TEST_std)	Encoder(TEST_std) (el Decoder ya no existe en esta etapa)	Encoder(TEST_std) (el Decoder ya no existe en esta etapa)
Escalado angular de TEST	MinMaxScaler.transform(Z_test) utilizando los mínimos y máximos calculados únicamente en TRAIN , obteniendo las 3 variables finales en el intervalo $[-\pi/2, \pi/2]$			
Salida final	TRAIN 3F y TEST 3F	TRAIN 3F y TEST 3F	TRAIN 3F y TEST 3F	TRAIN 3F y TEST 3F

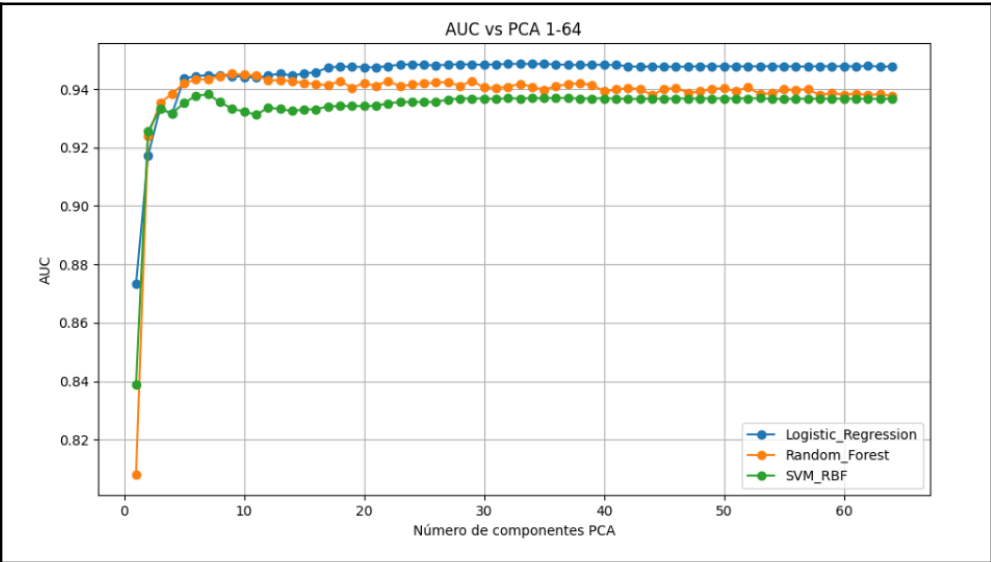


Error de reconstrucción de z64 para las 4 transformaciones

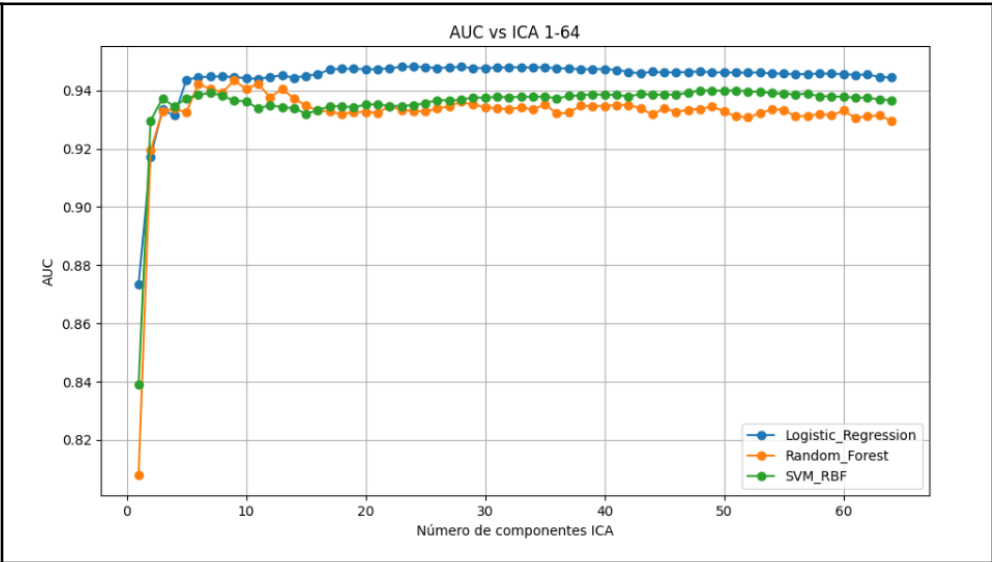


AUC de z64 para las 4 transformaciones (SVM RBF / RF / LR).

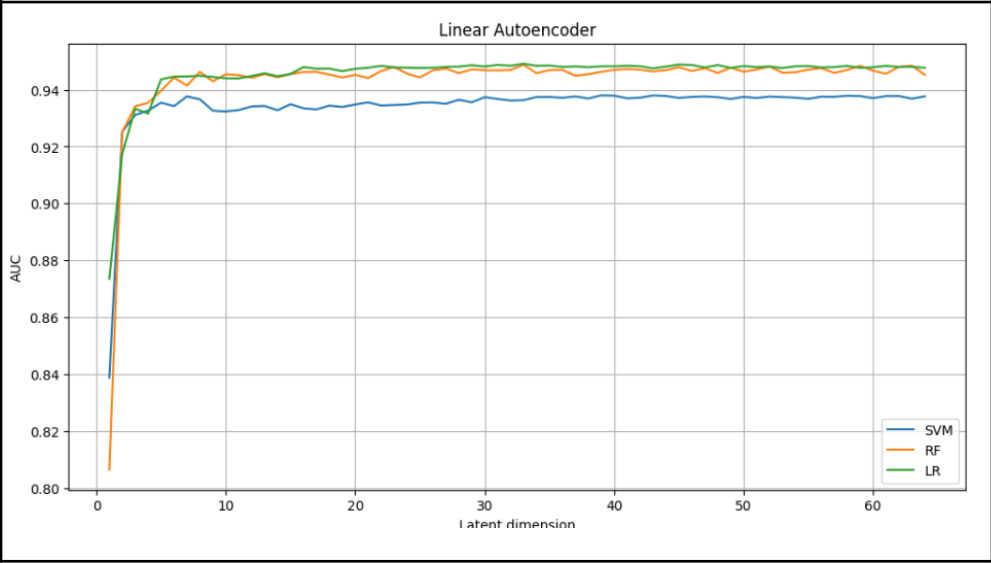
PCA



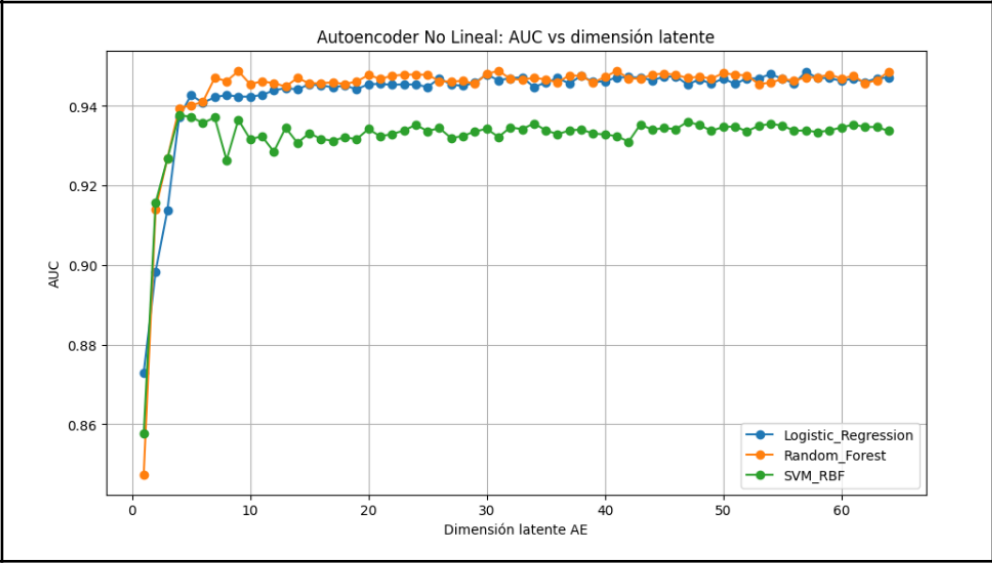
ICA



AE-Lineal



AE-No Lineal



Area Under the Curve



ACC de z64 para las 4 transformaciones (SVM RBF / RF / LR).

PCA

Model PCA Acc F1 bAcc AUC							Model PCA Acc F1 bAcc AUC							Model PCA Acc F1 bAcc AUC								
0	SVM_RBF	1	0.793454	0.789508	0.794085	0.838873	0	Random_Forest	1	0.702574	0.703422	0.702767	0.807961	0	Logistic_Regression	1	0.794407	0.786398	0.795390	0.873362		
1	SVM_RBF	2	0.872895	0.861496	0.874745	0.925457	1	Random_Forest	2	0.857960	0.853778	0.858798	0.923871	1	Logistic_Regression	2	0.872259	0.863729	0.873740	0.917171		
2	SVM_RBF	3	0.877344	0.866897	0.879124	0.933171	2	Random_Forest	3	0.877661	0.873646	0.878566	0.935234	2	Logistic_Regression	3	0.886876	0.879973	0.888267	0.933343		
3	SVM_RBF	4	0.882428	0.872940	0.884138	0.931693	3	Random_Forest	4	0.887512	0.883322	0.888497	0.938300	3	Logistic_Regression	4	0.886241	0.879380	0.887619	0.931453		
4	SVM_RBF	5	0.888465	0.879836	0.890124	0.935190	4	Random_Forest	5	0.886876	0.882119	0.887945	0.941993	4	Logistic_Regression	5	0.891325	0.884693	0.892717	0.943613		
...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...			
59	SVM_RBF	60	0.892596	0.885034	0.894144	0.936673	59	Random_Forest	60	0.876072	0.870690	0.877161	0.938014	59	Logistic_Regression	60	0.893232	0.887475	0.894505	0.947780		
60	SVM_RBF	61	0.892596	0.885034	0.894144	0.936656	60	Random_Forest	61	0.880521	0.875826	0.881540	0.938468	60	Logistic_Regression	61	0.893867	0.888220	0.895129	0.947766		
61	SVM_RBF	62	0.893232	0.885792	0.894767	0.936658	61	Random_Forest	62	0.878297	0.873388	0.879333	0.937903	61	Logistic_Regression	62	0.893549	0.887772	0.894829	0.947873		
62	SVM_RBF	63	0.893232	0.885792	0.894767	0.936611	62	Random_Forest	63	0.877661	0.872643	0.878709	0.938275	62	Logistic_Regression	63	0.892914	0.887178	0.894181	0.947712		
63	SVM_RBF	64	0.892914	0.885413	0.894456	0.936642	63	Random_Forest	64	0.879568	0.874628	0.880616	0.937566	63	Logistic_Regression	64	0.893549	0.887847	0.894817	0.947655		
54 rows × 6 columns							54 rows × 6 columns							64 rows × 6 columns								
Mejor SVM por AUC:							Mejor Random Forest por AUC:							Mejor Logistic Regression por AUC:								
Model	PCA	Acc	F1	bAcc	AUC		Model	PCA	Acc	F1	bAcc	AUC		Model	PCA	Acc	F1	bAcc	AUC			
6	SVM_RBF	7	0.888783	0.880383	0.890413	0.938154		8	Random_Forest	9	0.884652	0.878798	0.885869	0.945281		32	Logistic_Regression	33	0.896092	0.890379	0.897384	0.948609

AE-Lineal

AE						Acc						F1						bAcc						AUC					
0	1	0.793772	0.789900	0.794397	0.838734	0	1	0.706705	0.709108	0.706798	0.806431	0	1	0.794407	0.786539	0.795378	0.873493	0	1	0.794407	0.786539	0.795378	0.873493						
1	2	0.872259	0.860610	0.874133	0.925253	1	2	0.859231	0.854897	0.860093	0.925101	1	2	0.872259	0.863729	0.873740	0.917486	1	2	0.872259	0.863729	0.873740	0.917486						
2	3	0.878297	0.868249	0.880036	0.931095	2	3	0.876708	0.872787	0.877594	0.934171	2	3	0.886241	0.879299	0.887631	0.933315	2	3	0.886241	0.879299	0.887631	0.933315						
3	4	0.880521	0.870434	0.882291	0.932659	3	4	0.880203	0.875207	0.881264	0.935359	3	4	0.886876	0.879892	0.888279	0.931590	3	4	0.886876	0.879892	0.888279	0.931590						
4	5	0.887830	0.878985	0.889513	0.935422	4	5	0.886241	0.880826	0.887405	0.939642	4	5	0.891007	0.884317	0.892405	0.943669	4	5	0.891007	0.884317	0.892405	0.943669						
...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...						
59	60	0.894185	0.886927	0.895703	0.937029	59	60	0.893232	0.887625	0.894481	0.946655	59	60	0.893867	0.888145	0.895141	0.947863	59	60	0.893867	0.888145	0.895141	0.947863						
60	61	0.892596	0.885112	0.894132	0.937750	60	61	0.892914	0.887329	0.894158	0.945579	60	61	0.893867	0.887995	0.895165	0.948376	60	61	0.893867	0.887995	0.895165	0.948376						
61	62	0.892914	0.885491	0.894444	0.937768	61	62	0.894820	0.889850	0.895981	0.948028	61	62	0.893867	0.887995	0.895165	0.947970	61	62	0.893867	0.887995	0.895165	0.947970						
62	63	0.893549	0.886171	0.895079	0.936813	62	63	0.892914	0.887479	0.894134	0.948431	62	63	0.894503	0.888889	0.895765	0.948050	62	63	0.894503	0.888889	0.895765	0.948050						
63	64	0.892278	0.884733	0.893820	0.937659	63	64	0.892278	0.886356	0.893569	0.945198	63	64	0.895774	0.890154	0.897048	0.947674	63	64	0.895774	0.890154	0.897048	0.947674						
i4 rows × 5 columns						54 rows × 5 columns						64 rows × 5 columns																	
AE						Acc						F1						bAcc						AUC					
38	39	0.894185	0.886927	0.895703	0.937998	32	33	0.891961	0.88644	0.893186	0.948797	32	33	0.896727	0.890903	0.898044	0.949115	32	33	0.896727	0.890903	0.898044	0.949115						

ICA

Model ICA Acc F1 bAcc AUC						Model ICA Acc F1 bAcc AUC						Model ICA Acc F1 bAcc AUC								
0	SVM_RBF	1	0.793454	0.789508	0.794085	0.838940	0	Random_Forest	1	0.702256	0.703199	0.702443	0.807852	0	Logistic_Regression	1	0.794407	0.786398	0.795390	0.873361
1	SVM_RBF	2	0.871306	0.859424	0.873198	0.929592	1	Random_Forest	2	0.855418	0.850673	0.856315	0.919387	1	Logistic_Regression	2	0.872259	0.863636	0.873752	0.917171
2	SVM_RBF	3	0.878297	0.867885	0.880084	0.937327	2	Random_Forest	3	0.878615	0.873844	0.879633	0.932833	2	Logistic_Regression	3	0.887194	0.880189	0.888603	0.933411
3	SVM_RBF	4	0.881157	0.871478	0.882879	0.934436	3	Random_Forest	4	0.881474	0.876449	0.882547	0.933799	3	Logistic_Regression	4	0.885923	0.878757	0.887343	0.931411
4	SVM_RBF	5	0.889418	0.880985	0.891060	0.937381	4	Random_Forest	5	0.884652	0.879282	0.885797	0.932520	4	Logistic_Regression	5	0.891643	0.884914	0.893053	0.943691
...	...	...	...	...	...	...	59	Random_Forest	60	0.859549	0.859146	0.859905	0.933096	59	Logistic_Regression	60	0.891643	0.885762	0.892922	0.945601
59	SVM_RBF	60	0.885923	0.877766	0.887486	0.937842	60	Random_Forest	61	0.850651	0.850699	0.850945	0.930609	60	Logistic_Regression	61	0.891643	0.885762	0.892922	0.945361
60	SVM_RBF	61	0.885605	0.877551	0.887150	0.937625	61	Random_Forest	62	0.854465	0.853111	0.854938	0.931114	61	Logistic_Regression	62	0.892278	0.886356	0.893569	0.945491
61	SVM_RBF	62	0.885288	0.877253	0.886827	0.937515	62	Random_Forest	63	0.857960	0.857689	0.858297	0.931512	62	Logistic_Regression	63	0.891007	0.885169	0.892274	0.944671
62	SVM_RBF	63	0.884970	0.876871	0.886515	0.936983	63	Random_Forest	64	0.854147	0.853495	0.854531	0.929565	63	Logistic_Regression	64	0.891961	0.886059	0.893246	0.944471
63	SVM_RBF	64	0.885288	0.877253	0.886827	0.936675	Mejor Random Forest por AUC:						Mejor Logistic Regression por AUC:							
Model ICA Acc F1 bAcc AUC						Model ICA Acc F1 bAcc AUC														
8	Random Forest	9	0.883381	0.878114	0.884502	0.943453	23	Loistic Rearession	24	0.89482	0.889038	0.896112	0.948203							
54 rows × 6 columns																				
Mejor SVM por AUC:																				
Model	ICA	Acc	F1	bAcc	AUC															
50	SVM_RBF	51	0.885605	0.876881	0.887246	0.940075														

AE-No Lineal

Model AE Acc F1 bAcc AUC							Model AE Acc F1 bAcc AUC							Model AE Acc F1 bAcc AUC						
0	SVM_RBF	1	0.792819	0.776712	0.794450	0.857621	0	Random_Forest	1	0.742930	0.750540	0.742619	0.847361	0	Logistic_Regression	1	0.798221	0.775539	0.800396	0.872971
1	SVM_RBF	2	0.876072	0.864017	0.878043	0.915651	1	Random_Forest	2	0.843343	0.838414	0.844212	0.914024	1	Logistic_Regression	2	0.849698	0.839607	0.851177	0.898335
2	SVM_RBF	3	0.876072	0.864771	0.877948	0.926771	2	Random_Forest	3	0.873848	0.867975	0.874990	0.926841	2	Logistic_Regression	3	0.880839	0.873524	0.882233	0.913825
3	SVM_RBF	4	0.882745	0.873326	0.884450	0.937781	3	Random_Forest	4	0.881792	0.876658	0.882883	0.939491	3	Logistic_Regression	4	0.884652	0.878879	0.885857	0.937333
4	SVM_RBF	5	0.886241	0.877145	0.887941	0.937241	4	Random_Forest	5	0.883699	0.878888	0.884754	0.940181	4	Logistic_Regression	5	0.890372	0.883799	0.891746	0.942737
...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	
59	SVM_RBF	60	0.894820	0.887530	0.896351	0.934605	59	Random_Forest	60	0.891643	0.886295	0.892838	0.946894	59	Logistic_Regression	60	0.893232	0.887625	0.894481	0.946297
60	SVM_RBF	61	0.891007	0.883055	0.892596	0.935324	60	Random_Forest	61	0.895456	0.890370	0.896641	0.947571	60	Logistic_Regression	61	0.896092	0.890891	0.897301	0.946921
61	SVM_RBF	62	0.892914	0.885257	0.894479	0.934741	61	Random_Forest	62	0.893867	0.888667	0.895058	0.945754	61	Logistic_Regression	62	0.891325	0.885924	0.892526	0.946011
62	SVM_RBF	63	0.892596	0.884956	0.894156	0.934711	62	Random_Forest	63	0.891961	0.886667	0.893150	0.946511	62	Logistic_Regression	63	0.895138	0.889706	0.896377	0.946956
63	SVM_RBF	64	0.891007	0.883055	0.892596	0.933754	63	Random_Forest	64	0.891325	0.885772	0.892550	0.948641	63	Logistic_Regression	64	0.895456	0.890297	0.896653	0.947148

64 rows × 6 columns

Mejor SVM por AUC:

Model	AE	Acc	F1	bAcc	AUC	
3	SVM_RBF	4	0.882745	0.873326	0.884450	0.937781

34 rows × 6 columns

Mejor Random Forest por AUC:

Model	AE	Acc	F1	bAcc	AUC	
30	Random_Forest	31	0.893232	0.88785	0.894446	0.948889

64 rows × 6 columns

Mejor Logistic Regression por AUC:

Model	AE	Acc	F1	bAcc	AUC	
56	Logistic_Regression	57	0.896409	0.891478	0.897577	0.948581



Mejor resultado de z64 por modelo:(SVM RBF / RF / LR).

PCA

	Model	PCA	Acc	F1	bAcc	AUC
160	Logistic_Regression	33	0.896092	0.890379	0.897384	0.948609
72	Random_Forest	9	0.884652	0.878798	0.885869	0.945281
6	SVM_RBF	7	0.888783	0.880383	0.890413	0.938154

ICA

	Model	ICA	Acc	F1	bAcc	AUC
151	Logistic_Regression	24	0.894820	0.889038	0.896112	0.948203
72	Random_Forest	9	0.883381	0.878114	0.884502	0.943453
50	SVM_RBF	51	0.885605	0.876881	0.887246	0.940075

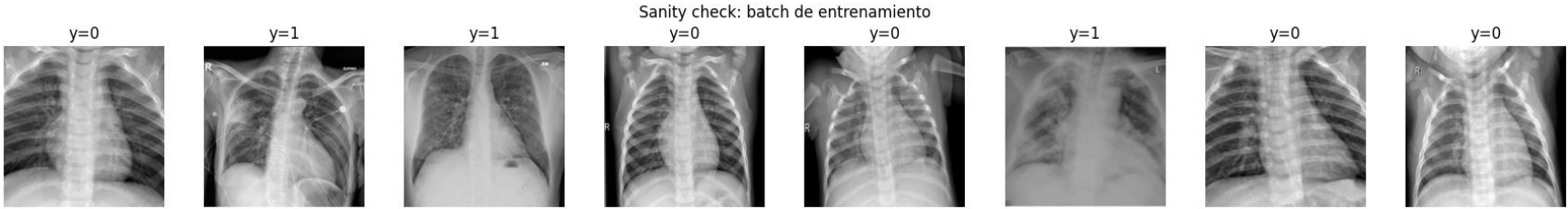
AE-Lineal

Model	AE	Acc	F1	bAcc	AUC
Logistic_Regression	32 33	0.896727	0.890903	0.898044	0.949115
Random_Forest	32 33	0.891961	0.88644	0.893186	0.948797
SVM_RBF	38 39	0.894185	0.886927	0.895703	0.937998

AE-No Lineal

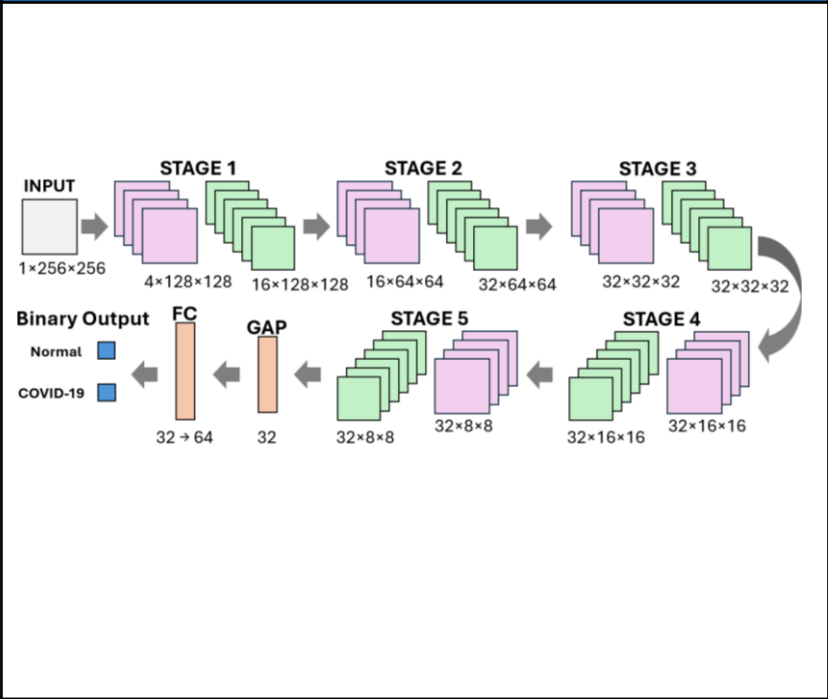
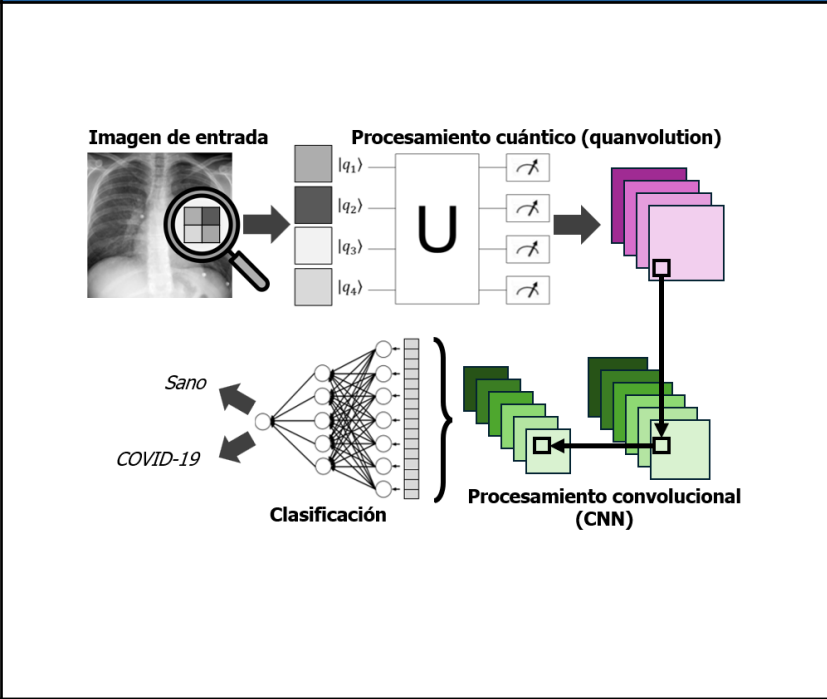
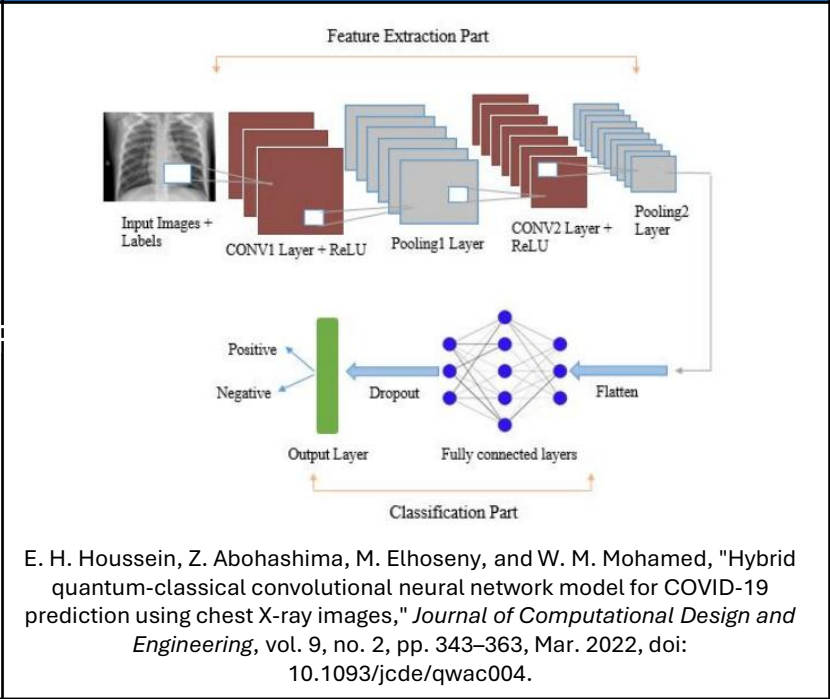
Model	AE	Acc	F1	bAcc	AUC
Random_Forest	31	0.893232	0.887850	0.894446	0.948889
Logistic_Regression	57	0.896409	0.891478	0.897577	0.948585
SVM_RBF	4	0.882745	0.873326	0.884450	0.937785

Aprendizaje de Representaciones Latentes en imágenes



- FASES**
- FASE 1:** EVALUACIÓN MODELO NORMAL.
 - FASE 2:** CONGELAR EXTRACTOR Y EXTRAER EMBEDDINGS z15.
 - FASE 3:** CLASIFICADOR AUXILIAR COMÚN SVM.

Las imágenes se redimensionaron a **256 × 256** píxeles y se almacenaron en formato **PNG**. El conjunto de datos se dividió en entrenamiento (**3642**), validación (**455**) y prueba (**455**), preservando la distribución de clases.

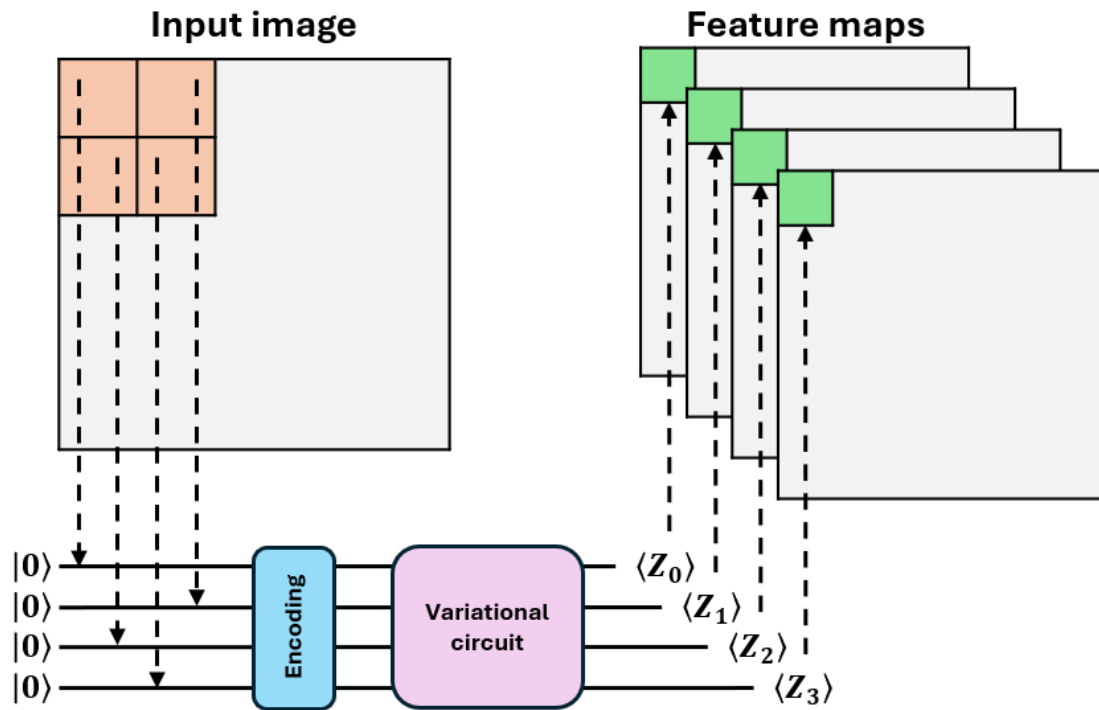
MS-HQCNN	HQCNN	CNN
 MS-HQCNN architecture diagram showing stages 1-5, FC, GAP, and binary output for Normal and COVID-19.	 HQCNN architecture diagram showing input image, quantum processing (U), and convolutional processing (CNN) for classification.	 CNN architecture diagram showing feature extraction part (CONV1, Pooling1, CONV2, Pooling2) and classification part (Dropout, Fully connected layers, Output Layer).

E. H. Houssein, Z. Abohashima, M. Elhoseny, and W. M. Mohamed, "Hybrid quantum-classical convolutional neural network model for COVID-19 prediction using chest X-ray images," *Journal of Computational Design and Engineering*, vol. 9, no. 2, pp. 343–363, Mar. 2022, doi: 10.1093/jcde/qvac004.



Quanvolution

La extracción de características se realiza mediante **quanvolution**, la imagen se divide en parches de 2×2 con stride 2 que son procesados de forma independiente, generando mapas de características que preservan la estructura espacial y sirven como entrada para las etapas posteriores de la red.



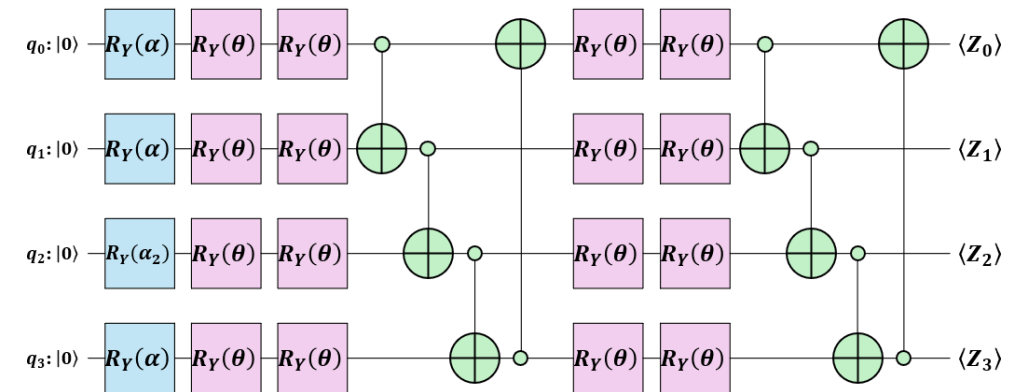
Cada parche se procesa con un **circuito cuántico parametrizado** de 4 qubits con codificación, rotaciones y entrelazamiento.

Encoding:
$$U_{enc}(\alpha) = \prod_{q=0}^3 R_Y(\alpha_q)$$

Ansatz:
$$V_\ell(\theta_\ell) = \left(\prod_{(i,j) \in \mathcal{R}} CNOT_{ij} \right) \left(\prod_{q=0}^3 R_Z(\theta_{\ell,q}^{(2)}) R_Y(\theta_{\ell,q}^{(1)}) \right)$$

Final state:
$$|\psi(\alpha, \theta)\rangle = \left(\prod_{\ell=1}^L V_\ell(\theta_\ell) \right) U_{enc}(\alpha) |0\rangle^{\otimes 4}$$

Measurement:
$$\langle Z_a \rangle, \text{ with } q \in \{0, 1, 2, 3\}$$

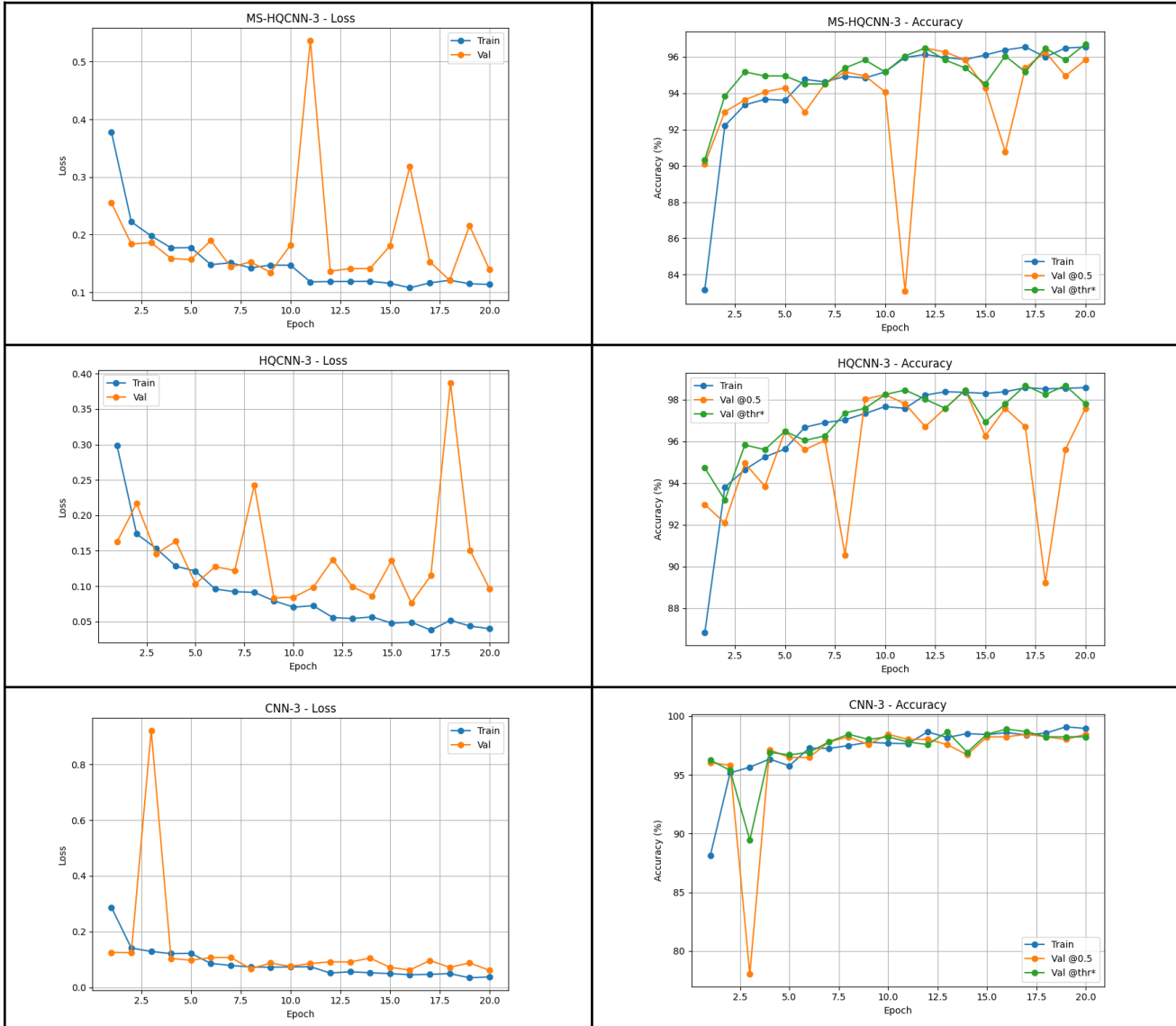


FASE 1: EVALUACIÓN MODELO NORMAL.

MS-HQCNN

HQCNN

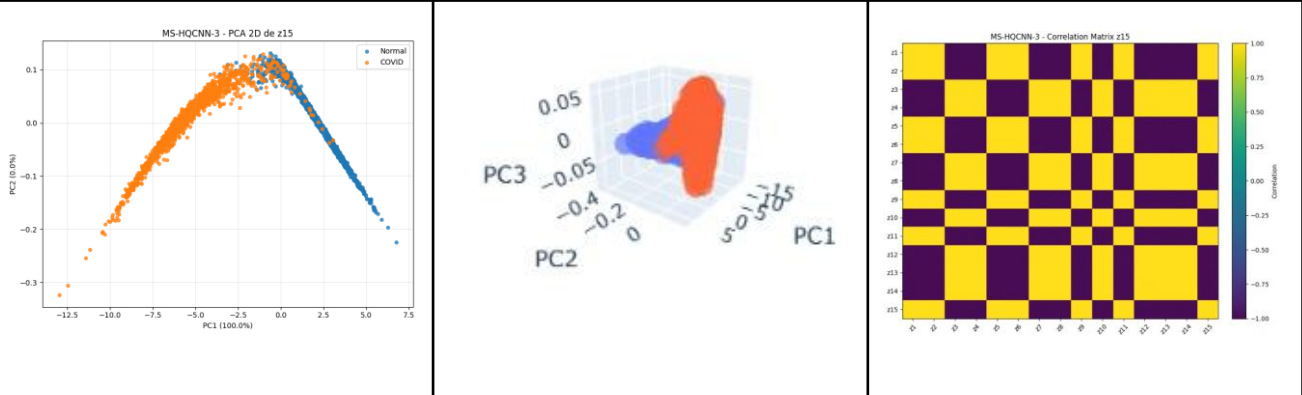
CNN



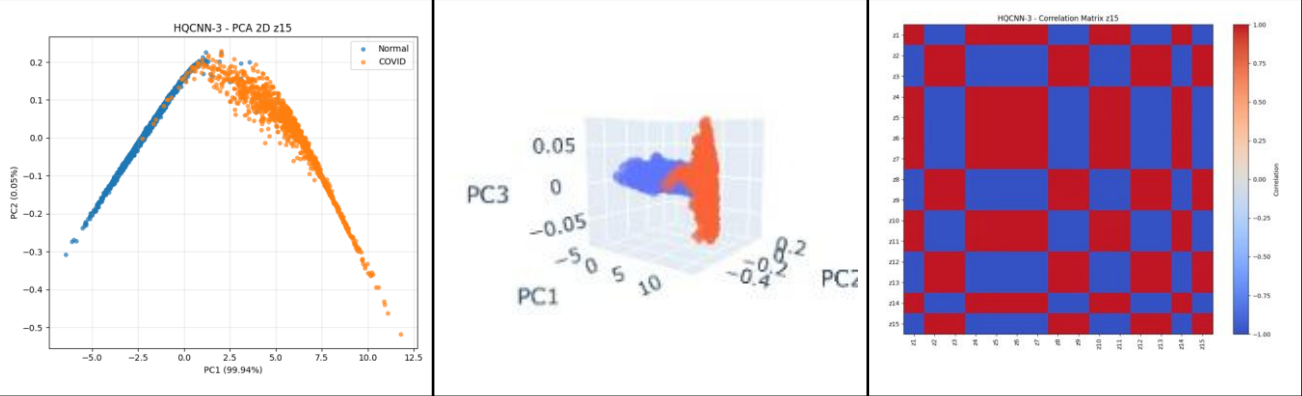
Métrica	MS-HQCNN	HQCNN	CNN
Loss	0.0771	0.1024	0.0398
AUC	0.9974	0.9984	0.9993
PR-AUC	0.9933	0.9964	0.9980
Accuracy (%)	97.37	98.68	99.34
Balanced Accuracy (%)	96.98	98.61	99.07
F1-score (%)	95.35	97.67	98.82
Precision (%)	94.62	96.92	99.21
Sensitivity (%)	96.09	98.44	98.44
Specificity (%)	97.87	98.78	99.70

FASE 2: CONGELAR EXTRACTOR Y EXTRAER EMBEDDINGS z15.

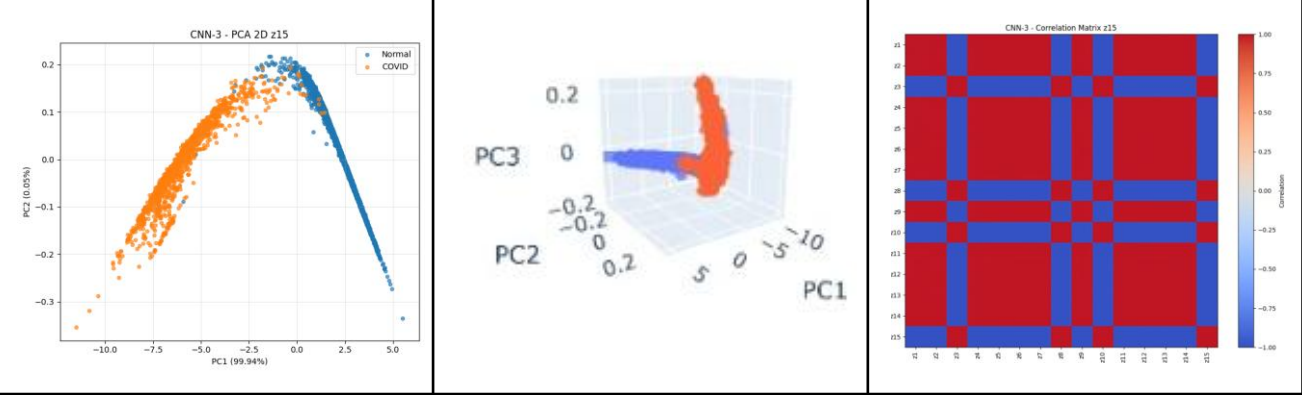
MS-HQCNN



HQCNN



CNN



FASE 3: CLASIFICADOR AUXILIAR COMÚN SVM.



Métrica	MS-HQCNN + SVM	HQCNN + SVM	CNN + SVM
Loss	—	—	—
AUC	0.9936	0.9903	0.9972
PR-AUC	0.9771	0.9840	0.9845
Accuracy (%)	98.03	99.12	99.34
Balanced Accuracy (%)	97.44	98.91	99.07
F1-score (%)	96.47	98.44	98.82
Precision (%)	96.85	98.44	99.21
Sensitivity (%)	96.09	98.44	98.44
Specificity (%)	98.78	99.39	99.70



Contenido:

- Experimento 1.

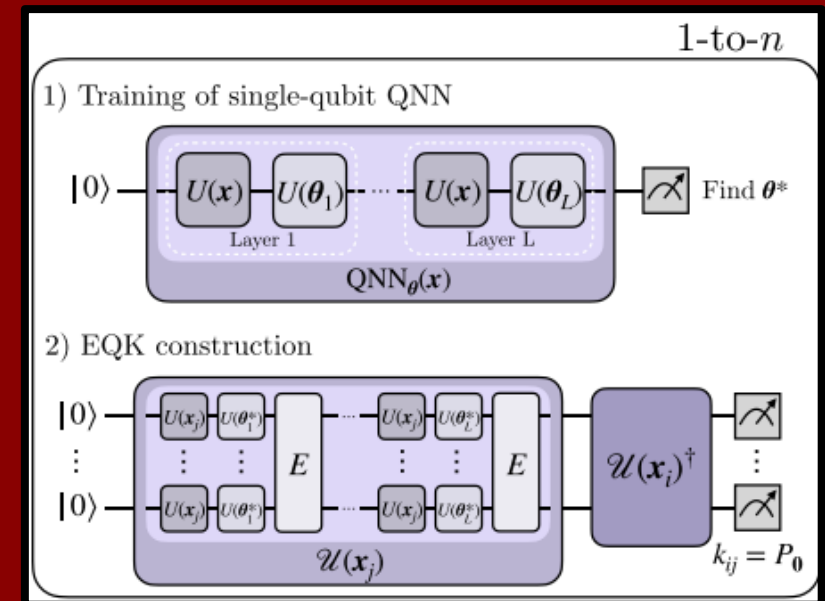
- **Experimento 2.**

Clasificación mediante NQK con reducción de dimensionalidad

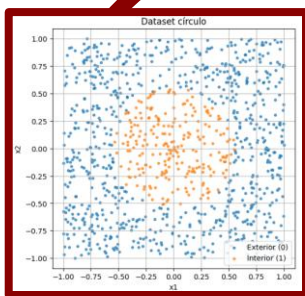
- Experimento 3.

- Experimento 4.

- Futuras pruebas.



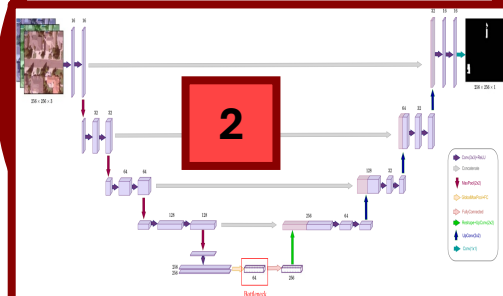
Clasificación mediante NQK con reducción de dimensionalidad



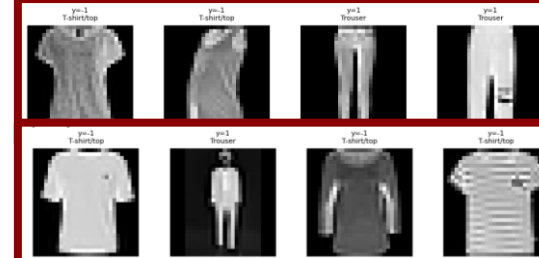
Data Re-uploading Quantum Neural Networks

https://github.com/Bayan0708/Quantum-Neural-Networks/tree/main/Data_Re-uploading_Quantum_Neural_Networks

Transformaciones DR QNN (Q1 a Q5)



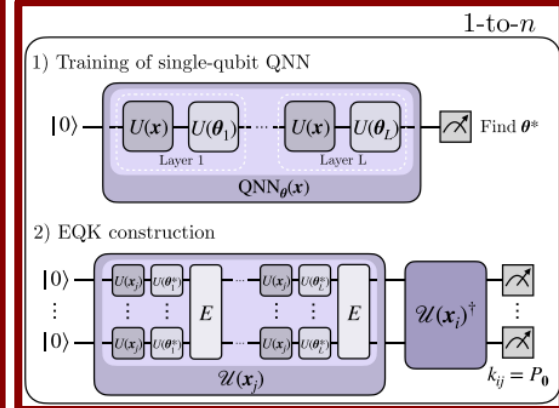
https://github.com/Bayan0708/Quantum-Neural-Networks/tree/main/DR_QNN_Batched_Q1_to_Q5



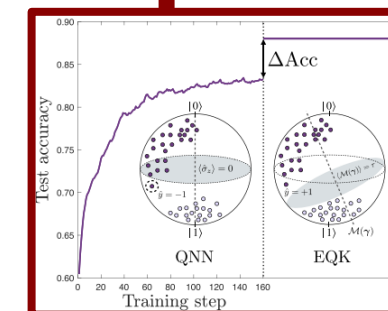
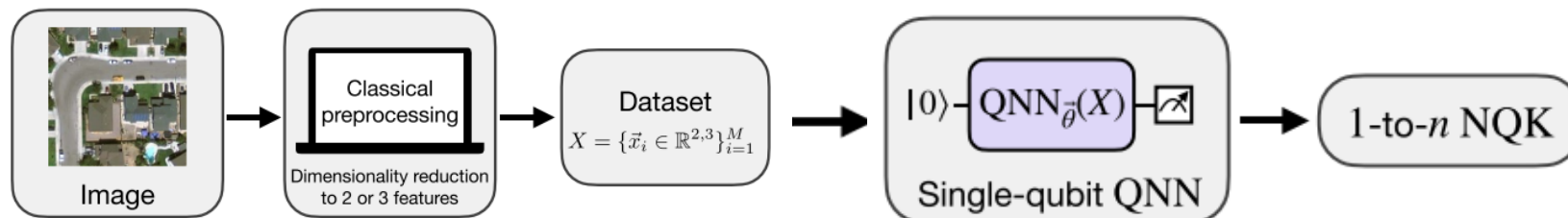
CNN DR QNN (SU(2) vs SU(4))

https://github.com/Bayan0708/Quantum-Neural-Networks/tree/main/Hybrid_Quantum_Deep_Learning

NQK con transformaciones de reducción a 3F



Obs: Solo se explica la cuarta etapa del avance.



P. Rodriguez-Grasa, Y. Ban,
and M. Sanz, "Neural
quantum kernels: Training
quantum kernels with
quantum neural networks,"
Physical Review Research,
vol. 7, no. 2, Art. no.
023269, Jun. 2025, doi:
10.1103/xphb-x2g4.

[https://github.com/Bayan0708/Quantum-Neural-Networks/tree/main/Neural Quantum Kernels](https://github.com/Bayan0708/Quantum-Neural-Networks/tree/main/Neural%20Quantum%20Kernels)



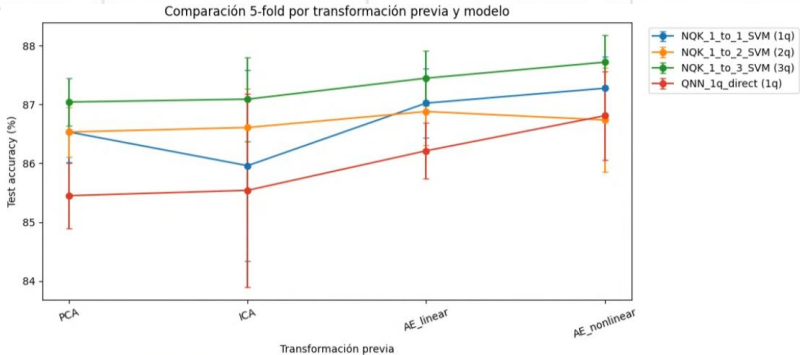
NQK con transformaciones de reducción a 3F

Resultados globales de todas las transformaciones (5 folds)

	transform	model	qubits	train_acc	test_acc	test_f1	test_auc	seconds
1	AE_linear	NQK_1_to_1_SVM	1	86.90 ± 0.61	87.02 ± 0.58	86.96 ± 0.58	0.9248 ± 0.0017	26.9 ± 1.1
2	AE_linear	NQK_1_to_2_SVM	2	86.73 ± 0.46	86.88 ± 0.57	86.80 ± 0.56	0.9167 ± 0.0071	28.0 ± 0.6
3	AE_linear	NQK_1_to_3_SVM	3	87.43 ± 0.29	87.45 ± 0.47	87.37 ± 0.47	0.9204 ± 0.0050	27.3 ± 0.3
4	AE_linear	QNN_1q_direct	1	86.35 ± 0.74	86.21 ± 0.48	86.13 ± 0.45	0.9264 ± 0.0048	1334.7 ± 70.9
5	AE_nonlinear	NQK_1_to_1_SVM	1	87.10 ± 0.46	87.28 ± 0.54	87.21 ± 0.54	0.9150 ± 0.0131	27.7 ± 0.7
6	AE_nonlinear	NQK_1_to_2_SVM	2	86.79 ± 0.84	86.74 ± 0.89	86.66 ± 0.87	0.9044 ± 0.0174	28.3 ± 0.6
7	AE_nonlinear	NQK_1_to_3_SVM	3	87.69 ± 0.19	87.72 ± 0.46	87.64 ± 0.48	0.9146 ± 0.0052	28.0 ± 0.9
8	AE_nonlinear	QNN_1q_direct	1	86.66 ± 0.69	86.81 ± 0.75	86.74 ± 0.74	0.9187 ± 0.0125	1355.7 ± 111.1
9	ICA	NQK_1_to_1_SVM	1	85.99 ± 1.40	85.96 ± 1.62	85.85 ± 1.67	0.9067 ± 0.0221	27.8 ± 0.9
10	ICA	NQK_1_to_2_SVM	2	86.66 ± 0.53	86.61 ± 0.66	86.51 ± 0.68	0.9125 ± 0.0080	28.4 ± 0.8
11	ICA	NQK_1_to_3_SVM	3	87.17 ± 0.55	87.09 ± 0.71	87.00 ± 0.73	0.9208 ± 0.0067	27.5 ± 0.7
12	ICA	QNN_1q_direct	1	85.41 ± 1.36	85.54 ± 1.65	85.37 ± 1.73	0.9095 ± 0.0247	1203.7 ± 187.6
13	PCA	NQK_1_to_1_SVM	1	86.56 ± 0.36	86.53 ± 0.52	86.45 ± 0.52	0.9199 ± 0.0031	28.5 ± 1.1
14	PCA	NQK_1_to_2_SVM	2	86.61 ± 0.19	86.53 ± 0.43	86.43 ± 0.44	0.9208 ± 0.0071	28.0 ± 0.5
15	PCA	NQK_1_to_3_SVM	3	87.16 ± 0.20	87.04 ± 0.40	86.94 ± 0.41	0.9241 ± 0.0082	27.8 ± 0.7
16	PCA	QNN_1q_direct	1	85.44 ± 0.73	85.45 ± 0.56	85.32 ± 0.60	0.9215 ± 0.0019	1238.8 ± 211.8

Comparación entre transformaciones previas, separada por modelo:

	model	qubits	AE_linear	AE_nonlinear	ICA	PCA
0	NQK_1_to_1_SVM	1	87.02 ± 0.58	87.28 ± 0.54	85.96 ± 1.62	86.53 ± 0.52
1	NQK_1_to_2_SVM	2	86.88 ± 0.57	86.74 ± 0.89	86.61 ± 0.66	86.53 ± 0.43
2	NQK_1_to_3_SVM	3	87.45 ± 0.47	87.72 ± 0.46	87.09 ± 0.71	87.04 ± 0.40
3	QNN_1q_direct	1	86.21 ± 0.48	86.81 ± 0.75	85.54 ± 1.65	85.45 ± 0.56



NQK con transformaciones de reducción a 3F

Resultados globales de 3 métodos clásicos (5 folds)

	transform	model	qubits	train_acc	test_acc	test_f1	test_auc	seconds
1	AE_linear	LogisticRegression	0	87.54 ± 0.11	87.55 ± 0.49	87.49 ± 0.49	0.9283 ± 0.0076	0.1 ± 0.0
2	AE_linear	RandomForest	0	100.00 ± 0.00	87.01 ± 0.42	86.99 ± 0.42	0.9268 ± 0.0050	2.2 ± 0.1
3	AE_linear	SVM_RBF	0	87.49 ± 0.16	87.41 ± 0.46	87.33 ± 0.46	0.9262 ± 0.0066	23.0 ± 0.2
4	AE_nonlinear	LogisticRegression	0	87.59 ± 0.34	87.54 ± 0.71	87.50 ± 0.71	0.9159 ± 0.0134	0.1 ± 0.0
5	AE_nonlinear	RandomForest	0	100.00 ± 0.00	87.59 ± 0.37	87.56 ± 0.37	0.9240 ± 0.0062	2.3 ± 0.1
6	AE_nonlinear	SVM_RBF	0	87.85 ± 0.09	87.87 ± 0.42	87.79 ± 0.43	0.9227 ± 0.0056	22.7 ± 0.4
7	ICA	LogisticRegression	0	87.57 ± 0.13	87.53 ± 0.48	87.48 ± 0.49	0.9284 ± 0.0076	0.1 ± 0.0
8	ICA	RandomForest	0	100.00 ± 0.00	87.06 ± 0.57	87.04 ± 0.57	0.9264 ± 0.0043	2.2 ± 0.1
9	ICA	SVM_RBF	0	87.53 ± 0.16	87.46 ± 0.41	87.37 ± 0.42	0.9305 ± 0.0070	23.6 ± 0.5
10	PCA	LogisticRegression	0	87.57 ± 0.13	87.53 ± 0.48	87.48 ± 0.48	0.9284 ± 0.0075	0.1 ± 0.1
11	PCA	RandomForest	0	100.00 ± 0.00	86.81 ± 0.49	86.79 ± 0.49	0.9263 ± 0.0047	2.1 ± 0.1
12	PCA	SVM_RBF	0	87.53 ± 0.16	87.46 ± 0.42	87.37 ± 0.42	0.9305 ± 0.0070	23.3 ± 0.2

Comparación clásica entre transformaciones previas:

	model	qubits	AE_linear	AE_nonlinear	ICA	PCA
0	LogisticRegression	0	87.55 ± 0.49	87.54 ± 0.71	87.53 ± 0.48	87.53 ± 0.48
1	RandomForest	0	87.01 ± 0.42	87.59 ± 0.37	87.06 ± 0.57	86.81 ± 0.49
2	SVM_RBF	0	87.41 ± 0.46	87.87 ± 0.42	87.46 ± 0.41	87.46 ± 0.42

Algoritmo	Kernel utilizado	Nombre habitual
SVM	Lineal	Linear SVM
SVM	Polinomial	Polynomial SVM
SVM	RBF (Gaussiano)	SVM_RBF
SVM	Sigmoid	Sigmoid SVM
SVM	Kernel precomputado	Precomputed Kernel SVM



NQK con transformaciones de reducción a 3F

	transform	model	qubits	train_acc	test_acc	test_f1	test_auc	seconds
1	AE_linear	LogisticRegression	0	87.54 ± 0.11	87.55 ± 0.49	87.49 ± 0.49	0.9283 ± 0.0076	0.1 ± 0.0
2	AE_linear	NQK_1_to_1_SVM	1	86.90 ± 0.61	87.02 ± 0.58	86.96 ± 0.58	0.9248 ± 0.0017	26.9 ± 1.1
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11	AE_nonlinear	NQK_1_to_3_SVM	3	87.69 ± 0.19	87.72 ± 0.46	87.64 ± 0.48	0.9146 ± 0.0052	28.0 ± 0.9
12	AE_nonlinear	QNN_1q_direct	1	86.66 ± 0.69	86.81 ± 0.75	86.74 ± 0.74	0.9187 ± 0.0125	1355.7 ± 111.1
13	AE_nonlinear	RandomForest	0	100.00 ± 0.00	87.59 ± 0.37	87.56 ± 0.37	0.9240 ± 0.0062	2.3 ± 0.1
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15	ICA	LogisticRegression	0	87.57 ± 0.13	87.53 ± 0.48	87.48 ± 0.49	0.9284 ± 0.0076	0.1 ± 0.0
16	ICA	NQK_1_to_1_SVM	1	85.99 ± 1.40	85.96 ± 1.62	85.85 ± 1.67	0.9067 ± 0.0221	27.8 ± 0.9
17	ICA	NQK_1_to_2_SVM	2	86.66 ± 0.53	86.61 ± 0.66	86.51 ± 0.68	0.9125 ± 0.0080	28.4 ± 0.8
18	ICA	NQK_1_to_3_SVM	3	87.17 ± 0.55	87.09 ± 0.71	87.00 ± 0.73	0.9208 ± 0.0067	27.5 ± 0.7
19	ICA	QNN_1q_direct	1	85.41 ± 1.36	85.54 ± 1.65	85.37 ± 1.73	0.9095 ± 0.0247	1203.7 ± 187.6
20	ICA	RandomForest	0	100.00 ± 0.00	87.06 ± 0.57	87.04 ± 0.57	0.9264 ± 0.0043	2.2 ± 0.1
21	ICA	SVM_RBF	0	87.53 ± 0.16	87.46 ± 0.41	87.37 ± 0.42	0.9305 ± 0.0070	23.6 ± 0.5
22	PCA	LogisticRegression	0	87.57 ± 0.13	87.53 ± 0.48	87.48 ± 0.48	0.9284 ± 0.0075	0.1 ± 0.1
23	PCA	NQK_1_to_1_SVM	1	86.56 ± 0.36	86.53 ± 0.52	86.45 ± 0.52	0.9199 ± 0.0031	28.5 ± 1.1
24	PCA	NQK_1_to_2_SVM	2	86.61 ± 0.19	86.53 ± 0.43	86.43 ± 0.44	0.9208 ± 0.0071	28.0 ± 0.5
25	PCA	NQK_1_to_3_SVM	3	87.16 ± 0.20	87.04 ± 0.40	86.94 ± 0.41	0.9241 ± 0.0082	27.8 ± 0.7
26	PCA	QNN_1q_direct	1	85.44 ± 0.73	85.45 ± 0.56	85.32 ± 0.60	0.9215 ± 0.0019	1238.8 ± 211.8
27	PCA	RandomForest	0	100.00 ± 0.00	86.81 ± 0.49	86.79 ± 0.49	0.9263 ± 0.0047	2.1 ± 0.1
28	PCA	SVM_RBF	0	87.53 ± 0.16	87.46 ± 0.42	87.37 ± 0.42	0.9305 ± 0.0070	23.3 ± 0.2

1-to-n NQK

Model	# features (p)	Training accuracy	Test accuracy
SVC	$p = 2$	86.1±0.3	86.2±2.5
	$p = 3$	88.0±0.3	88.1±2.7
	$p = 45$	89.4±0.3	89.4±2.5
Random forest	$p = 2$	91.1±0.3	87.2±2.7
	$p = 3$	93.0±0.3	87.6±2.0
	$p = 10$	96.9±0.2	88.3±2.8
1-to-2 NQK	$p = 2$	86.6±0.3	86.5±2.5
	$p = 3$	81.7±1.8	81.1±2.9
1-to-3 NQK	$p = 2$	86.6±0.3	86.5±2.6
	$p = 3$	85.9±0.6	85.6±2.7

*Satellite image
classification with neural
quantum kernels*



Contenido:

- Experimento 1.
- Experimento 2.
- **Experimento 3.**

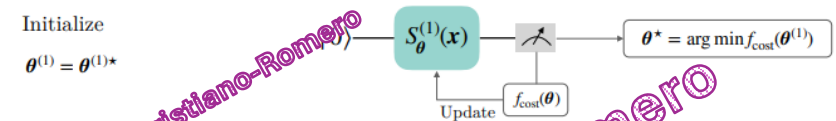
Clasificación mediante DR QNN SU(4) de Camila con reducción de dimensionalidad

- Experimento 4.
- Futuras pruebas.

1) Train a 1-qubit QNN

Initialize

$$\theta^{(1)} = \theta^{(1)*}$$



2) Add a second qubit and initialize using previous parameters

Initialize

$$\theta^{(2)} = \varphi^{(1-2)} = 0$$

$$\theta^{(1)} = \text{rand}(-1, 1)$$



3) Add a third qubit and initialize using previous parameters

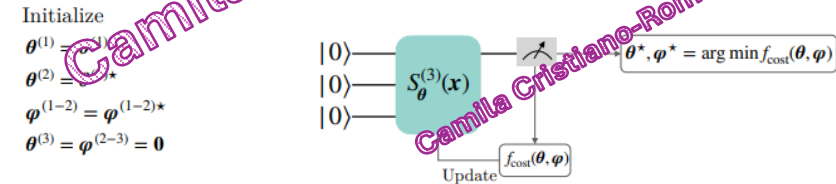
Initialize

$$\theta^{(1)} = \theta^{(1)*}$$

$$\theta^{(2)} = \theta^{(2)*}$$

$$\varphi^{(1-2)} = \varphi^{(1-2)*}$$

$$\theta^{(3)} = \varphi^{(2-3)} = 0$$

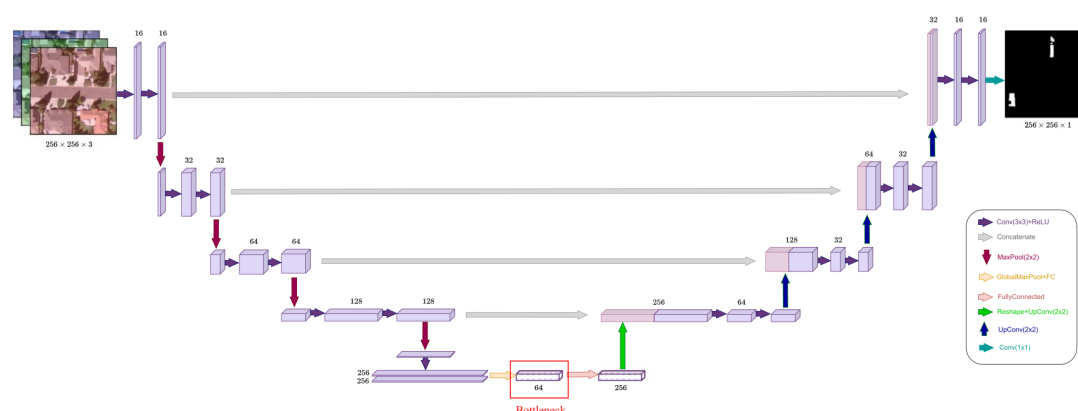


Up to n



Clasificación mediante DR QNN SU(4) de Camila con reducción de dimensionalidad

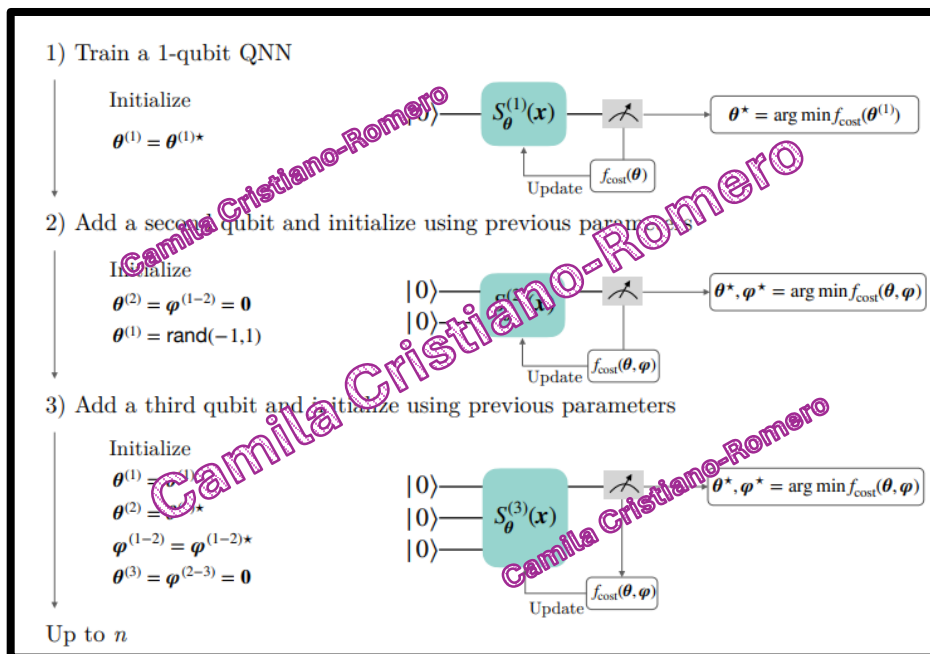
Pre procesado de datos // Transformaciones satelitales
de 1 a 15 características



- PCA
- ICA
- AE Lineal
- AE No lineal

- 60 CSV (15 x 4).
- 60 / 20 / 20.
- Configuración explicada arriba.
- # 20 976
- 5 folds

Clasificación



<https://drive.google.com/drive/folders/17m5QfvGxJkbpuBNov3d8NkwIAjhguYp?usp=sharing>

https://github.com/Bayan0708/Quantum-Neural-Networks/blob/main/Zn_Latent_Space_Analysis%20directory/Satellite_Transformacione_s_1_a_15_60_20_20.ipynb

Rendimiento general

Transformación	Qubits	Train Acc (%)	Val Acc (%)	Test Acc (%)	Test AUC	F1 macro
PCA	2	83.71 ± 0.69	83.62 ± 0.77	83.61 ± 0.78	0.9031 ± 0.0049	83.54
PCA	4	84.99 ± 0.56	84.99 ± 0.50	84.98 ± 0.51	0.9126 ± 0.0055	84.91
PCA	6	85.56 ± 0.41	85.48 ± 0.39	85.49 ± 0.38	0.9169 ± 0.0054	85.42
ICA	2	82.19 ± 3.15	82.11 ± 3.22	82.13 ± 3.23	0.8874 ± 0.0306	82.05
ICA	4	83.95 ± 2.76	83.87 ± 2.77	83.86 ± 2.77	0.9044 ± 0.0228	83.80
ICA	6	84.47 ± 2.47	84.42 ± 2.47	84.44 ± 2.49	0.9085 ± 0.0208	84.39
AE lineal	2	87.93 ± 0.61	87.89 ± 0.62	87.87 ± 0.62	0.9357 ± 0.0068	87.83
AE lineal	4	88.28 ± 0.79	88.21 ± 0.79	88.18 ± 0.79	0.9374 ± 0.0073	88.14
AE lineal	6	88.50 ± 0.69	88.39 ± 0.71	88.30 ± 0.70	0.9384 ± 0.0071	88.26
AE no lineal	2	87.96 ± 0.46	87.97 ± 0.46	87.94 ± 0.46	0.9352 ± 0.0046	87.90
AE no lineal	4	88.35 ± 0.46	88.34 ± 0.46	88.42 ± 0.47	0.9385 ± 0.0045	88.40
AE no lineal	6	88.42 ± 0.46	88.36 ± 0.47	88.35 ± 0.47	0.9389 ± 0.0047	88.32

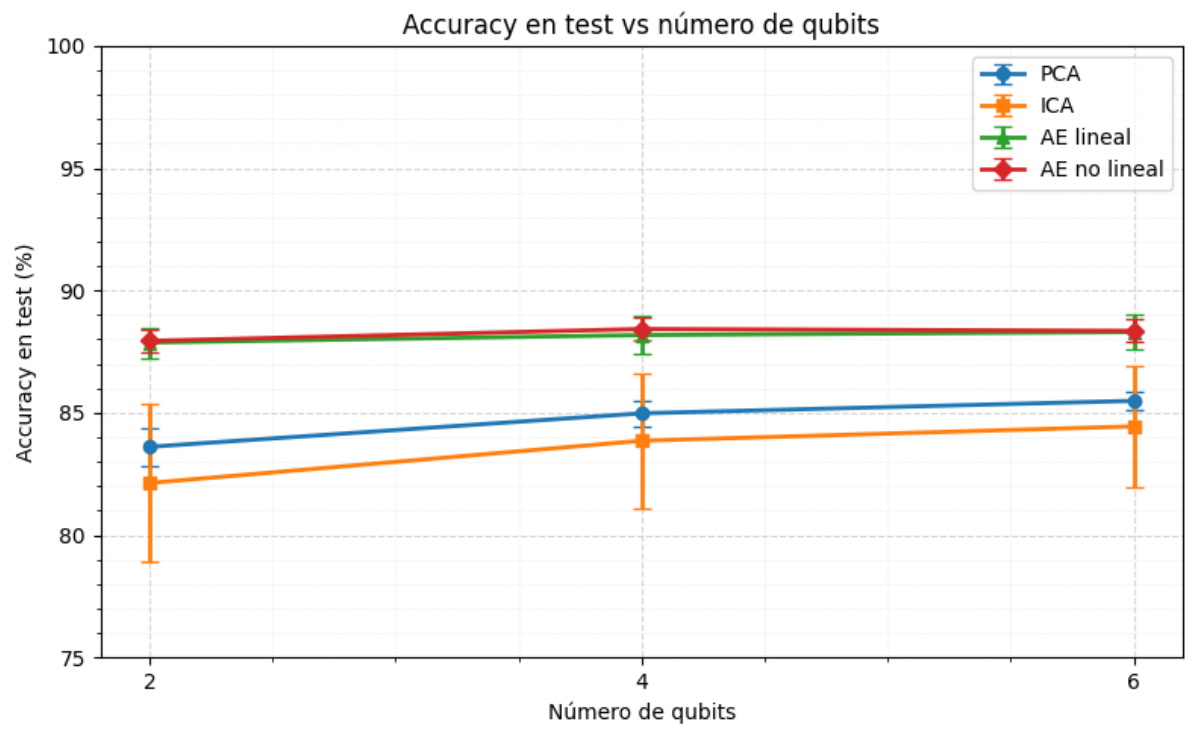
Comparativa con los resultados de Pablo

Transformación	Qubits	Train Acc (%)	Test Acc (%)
PCA	2	83.71 ± 0.69	83.61 ± 0.78
PCA	4	84.99 ± 0.56	84.98 ± 0.51
PCA	6	85.56 ± 0.41	85.49 ± 0.38
ICA	2	82.19 ± 3.15	82.13 ± 3.23
ICA	4	83.95 ± 2.76	83.86 ± 2.77
ICA	6	84.47 ± 2.47	84.44 ± 2.49
AE lineal	2	87.93 ± 0.61	87.87 ± 0.62
AE lineal	4	88.28 ± 0.79	88.18 ± 0.79
AE lineal	6	88.50 ± 0.69	88.30 ± 0.70
AE no lineal	2	87.96 ± 0.46	87.94 ± 0.46
AE no lineal	4	88.35 ± 0.46	88.42 ± 0.47
AE no lineal	6	88.42 ± 0.46	88.35 ± 0.47

Model	# features (p)	Training accuracy	Test accuracy
SVC	$p=2$	86.1±0.3	86.2±2.5
	$p=3$	88.0±0.3	88.1±2.7
	$p=45$	89.4±0.3	89.4±2.5
Random forest	$p=2$	91.1±0.3	87.2±2.7
	$p=3$	93.0±0.3	87.6±2.0
	$p=10$	96.9±0.2	88.3±2.8
1-to-2 NQK	$p=2$	86.6±0.3	86.5±2.5
	$p=3$	81.7±1.8	81.1±2.9
1-to-3 NQK	$p=2$	86.6±0.3	86.5±2.6
	$p=3$	85.9±0.6	85.6±2.7

¿Cuánto mejora al aumentar capas?

Transformación	2→4	4→6	2→6
PCA	+1.37 %	+0.51 %	+1.88 %
ICA	+1.73 %	+0.58 %	+2.31 %
AE lineal	+0.31 %	+0.12 %	+0.43 %
AE no lineal	+0.48 %	-0.07 %	+0.41 %



Estabilidad

Transformación	Mejor std (Test Acc)
PCA	±0.38
AE no lineal	±0.46
AE lineal	±0.70
ICA	±2.49

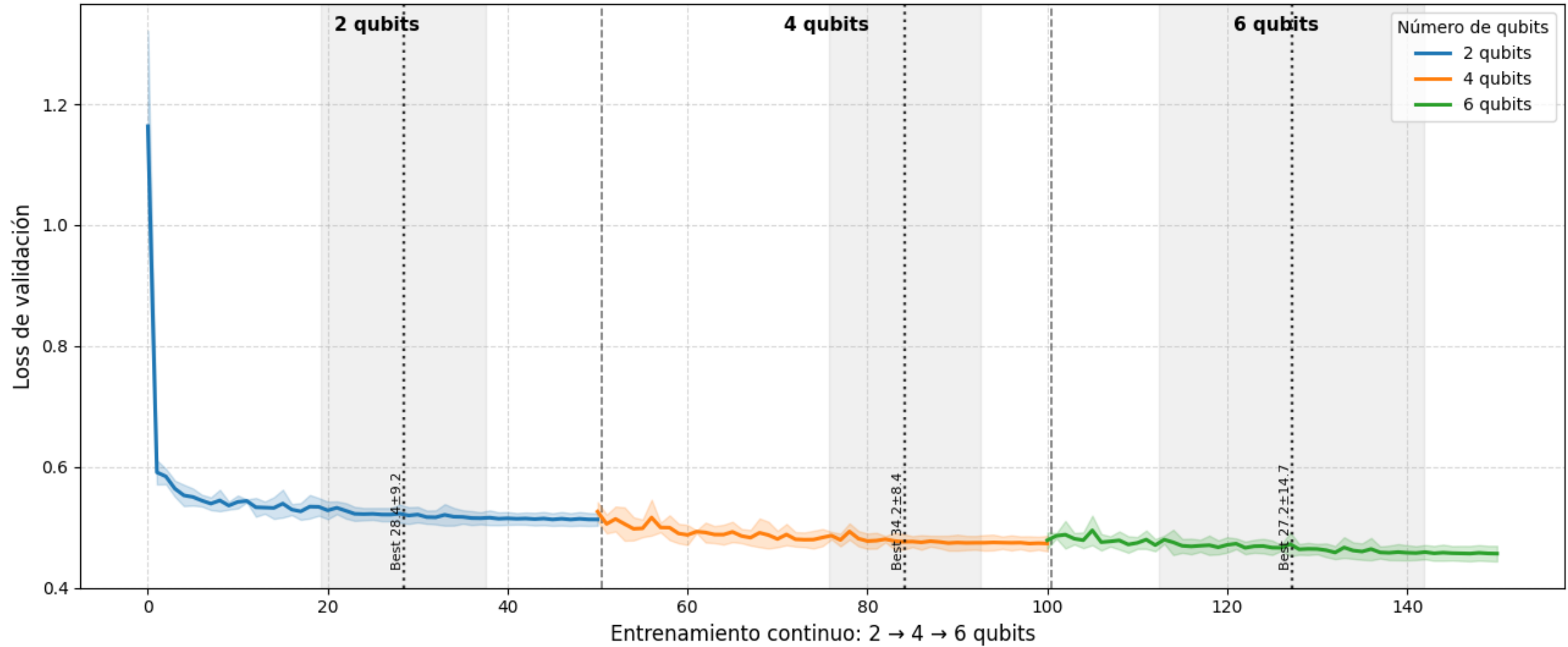
ICA no solo obtiene menor rendimiento, sino que además presenta una variabilidad muy superior entre folds.

Tiempo de entrenamiento

Transformación	2q (s)	4q (s)	6q (s)
PCA	446.0 ± 9.6	536.6 ± 5.2	718.9 ± 7.5
ICA	451.6 ± 6.8	546.0 ± 3.2	724.4 ± 1.5
AE lineal	451.1 ± 6.0	541.7 ± 1.9	726.6 ± 2.6
AE no lineal	452.3 ± 8.5	545.7 ± 5.2	726.8 ± 5.8

PCA

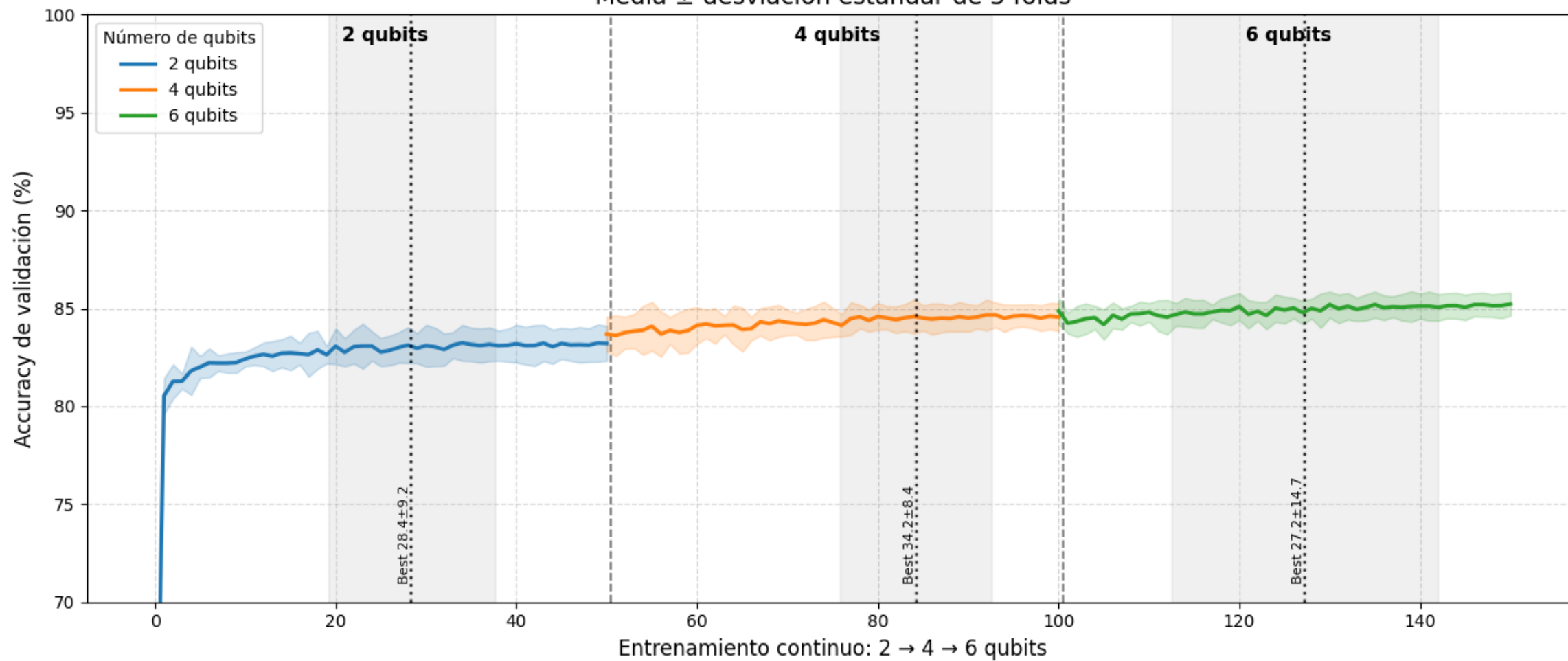
Loss de validación — PCA
Media $\pm$ desviación estándar de 5 folds





PCA

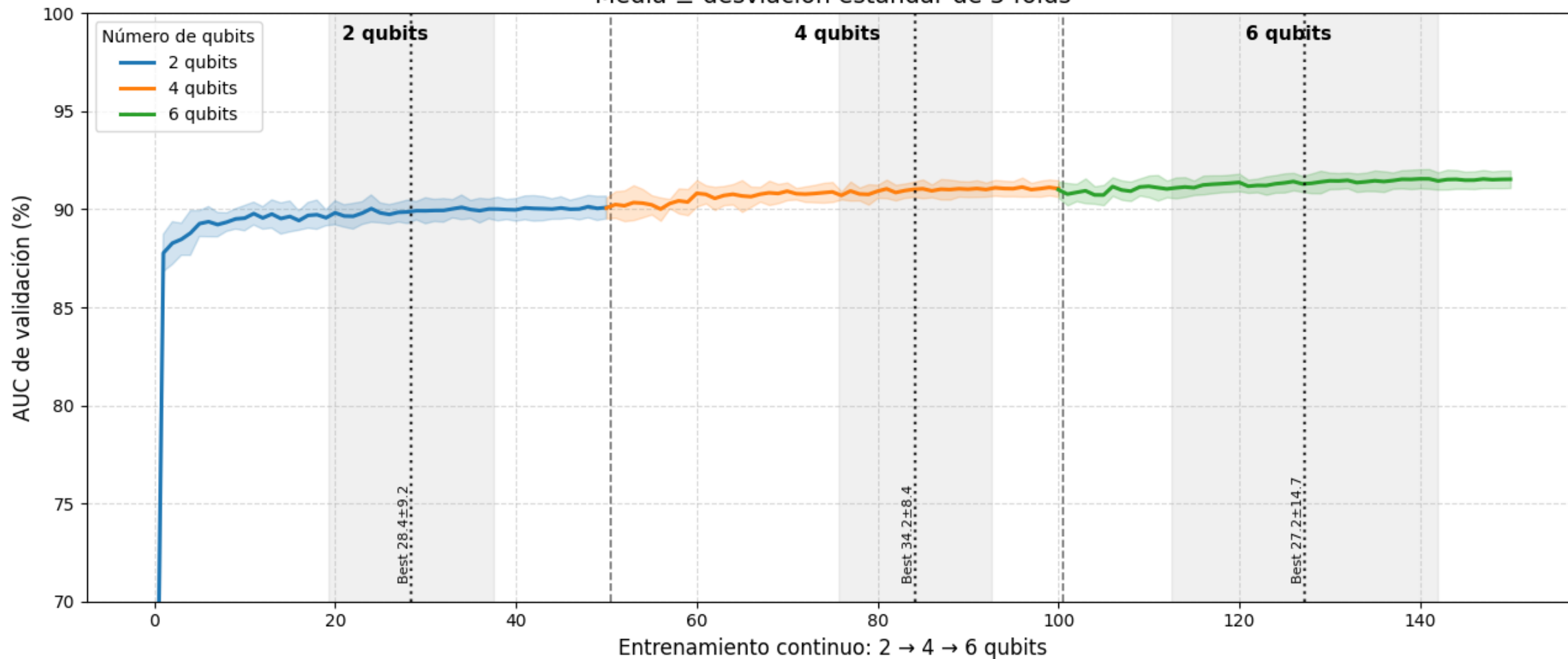
Accuracy de validación (%) — PCA
Media $\pm$ desviación estándar de 5 folds





PCA

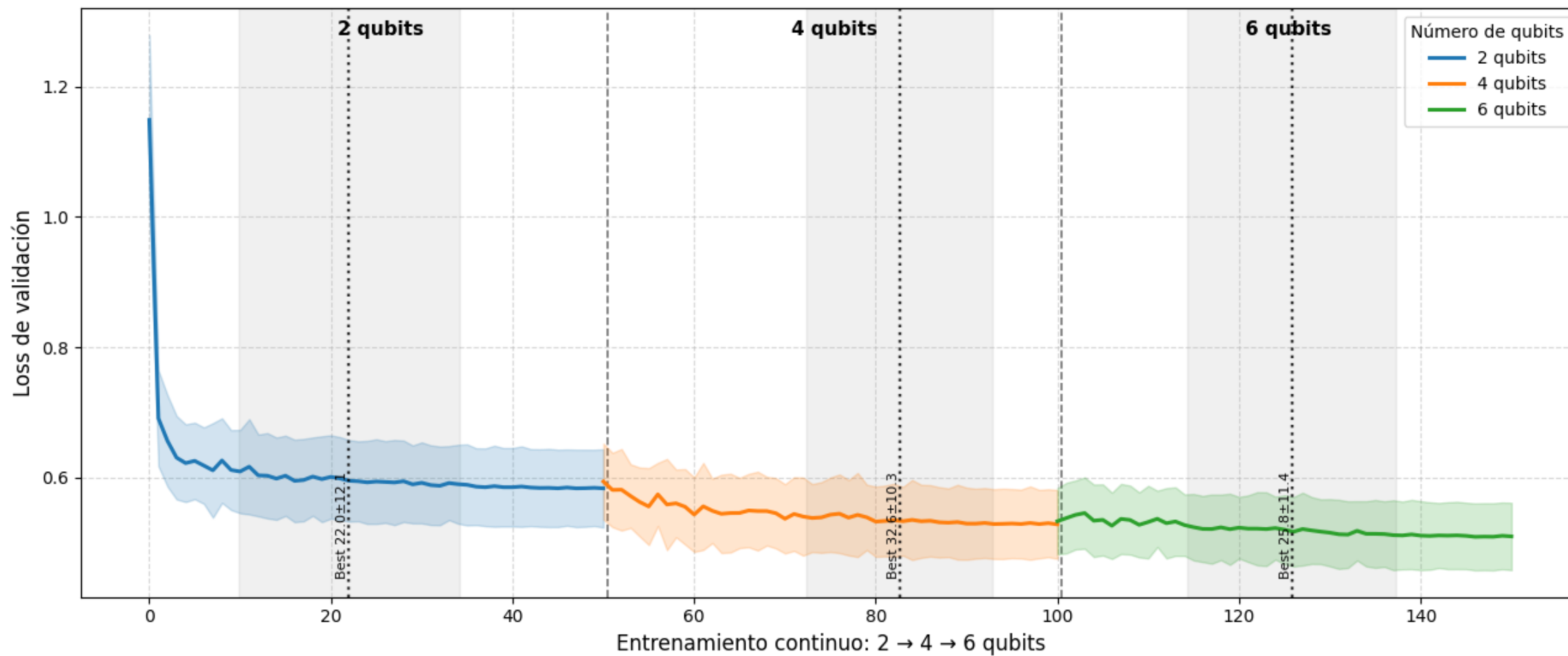
AUC de validación (%) — PCA
Media $\pm$ desviación estándar de 5 folds





ICA

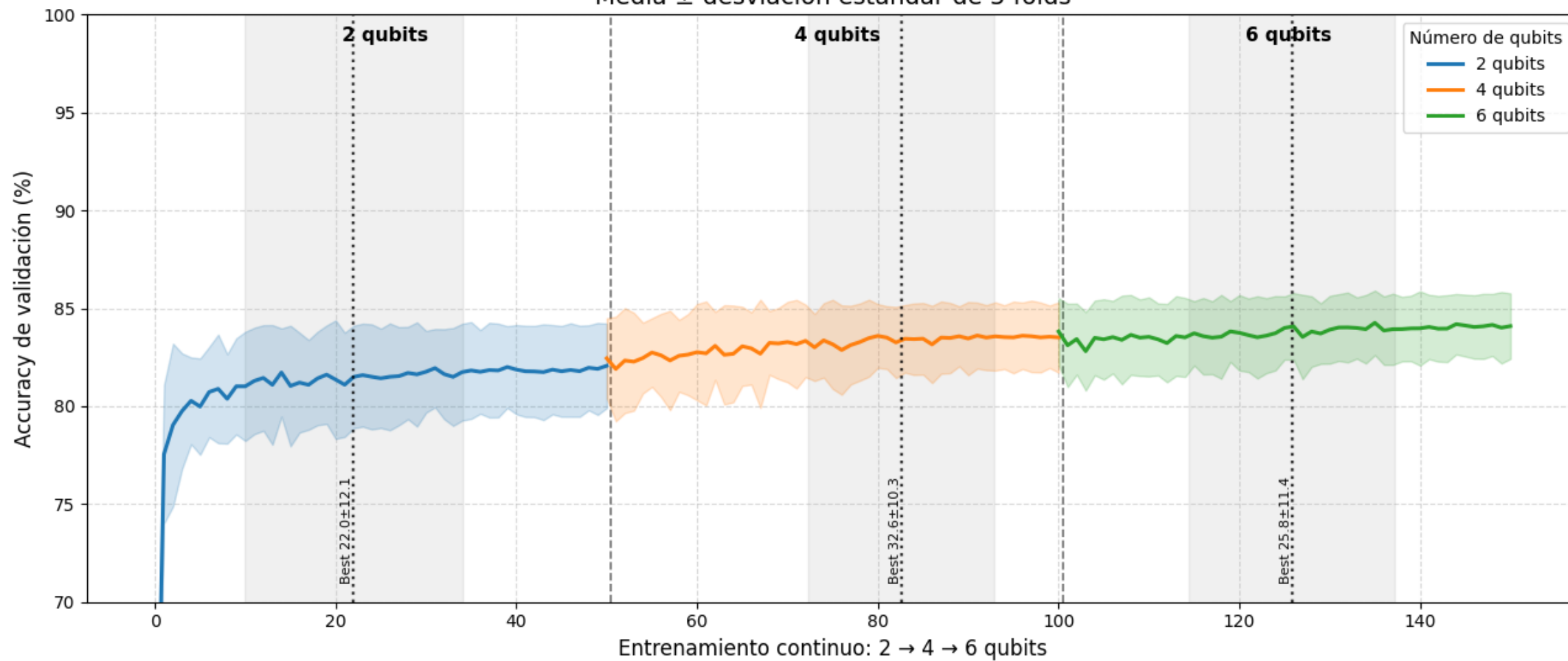
Loss de validación — ICA
Media $\pm$ desviación estándar de 5 folds





ICA

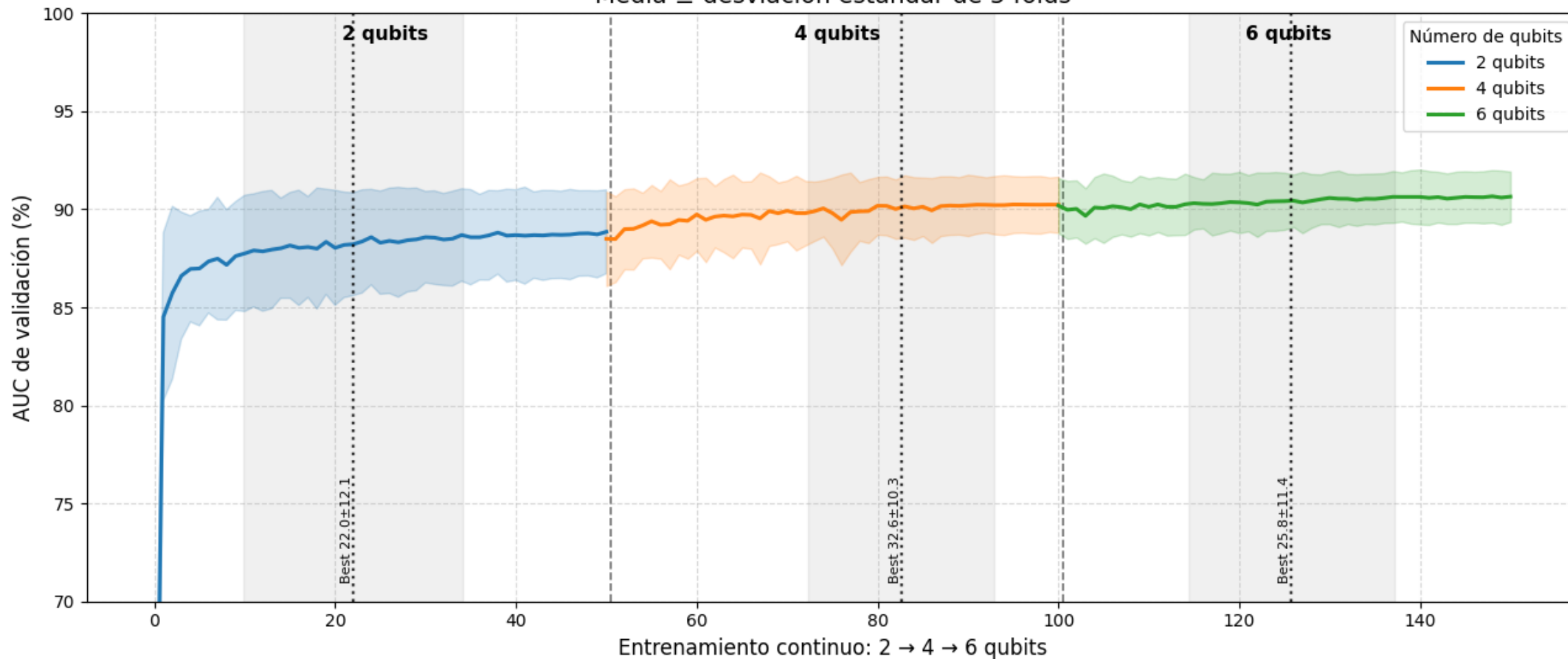
Accuracy de validación (%) — ICA
Media $\pm$ desviación estándar de 5 folds





ICA

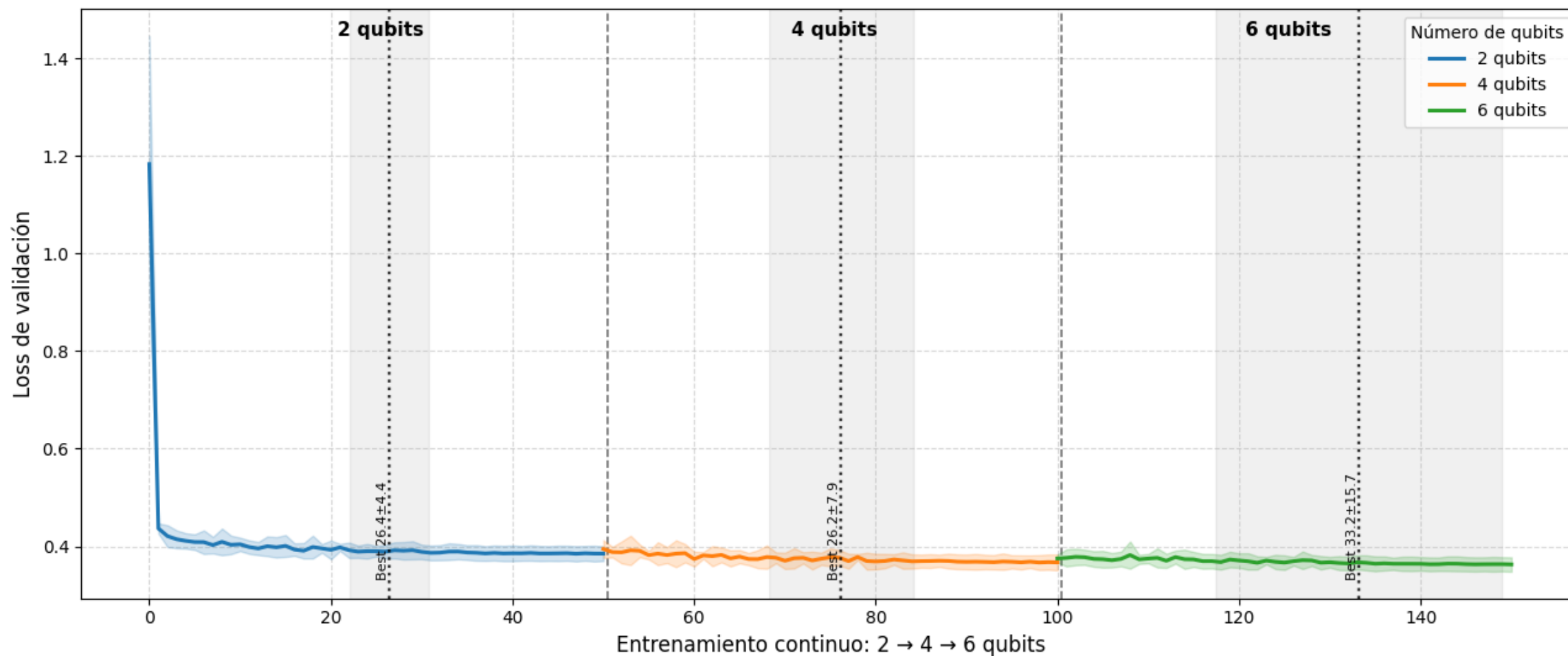
AUC de validación (%) — ICA
Media $\pm$ desviación estándar de 5 folds





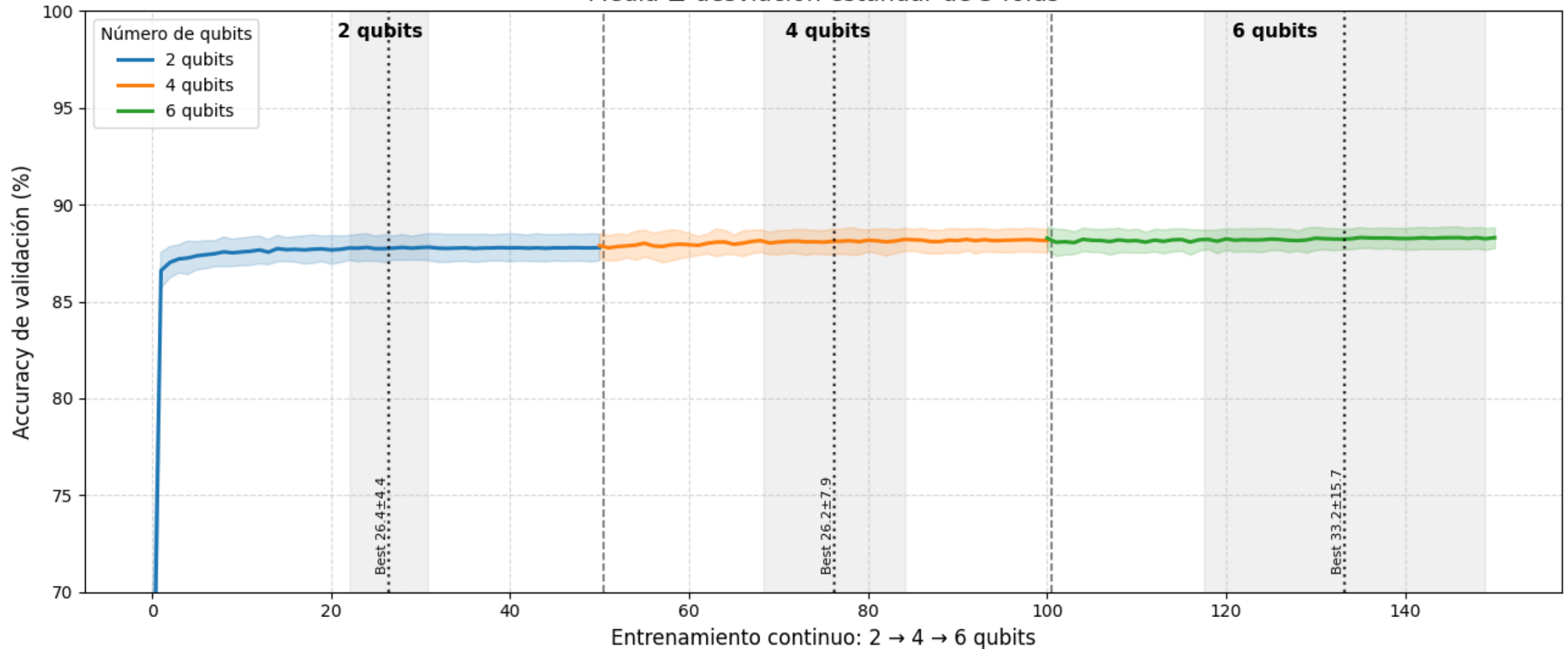
AE lineal

Loss de validación — AE lineal
Media $\pm$ desviación estándar de 5 folds



AE lineal

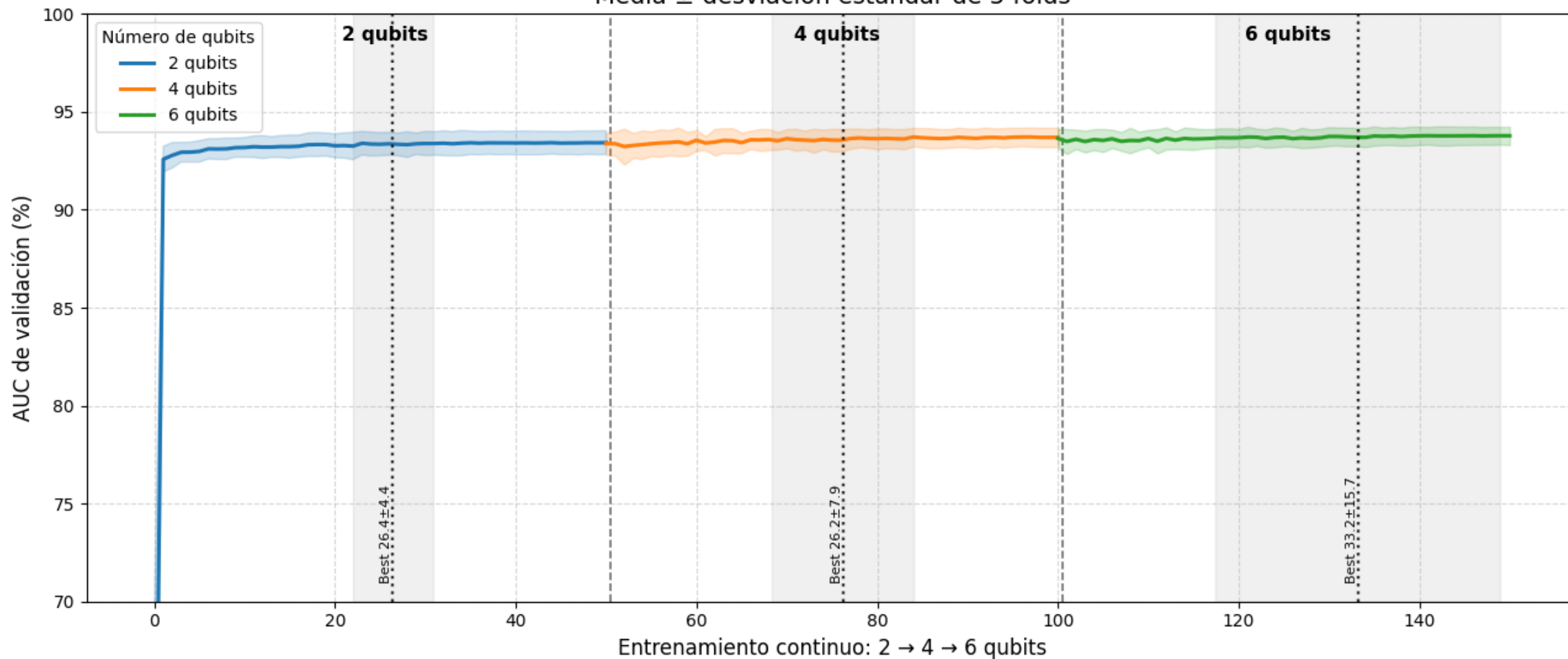
Accuracy de validación (%) — AE lineal
Media $\pm$ desviación estándar de 5 folds





AE lineal

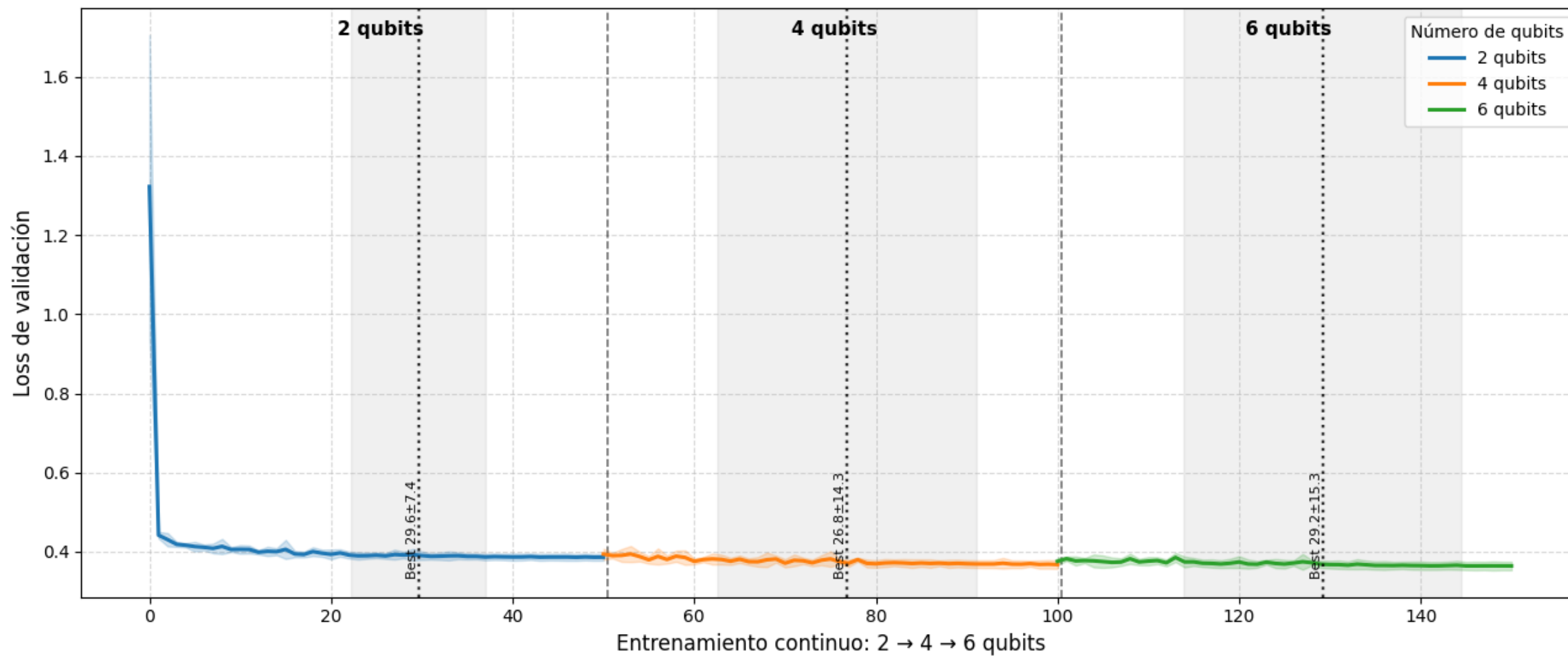
AUC de validación (%) — AE lineal
Media $\pm$ desviación estándar de 5 folds





AE no lineal

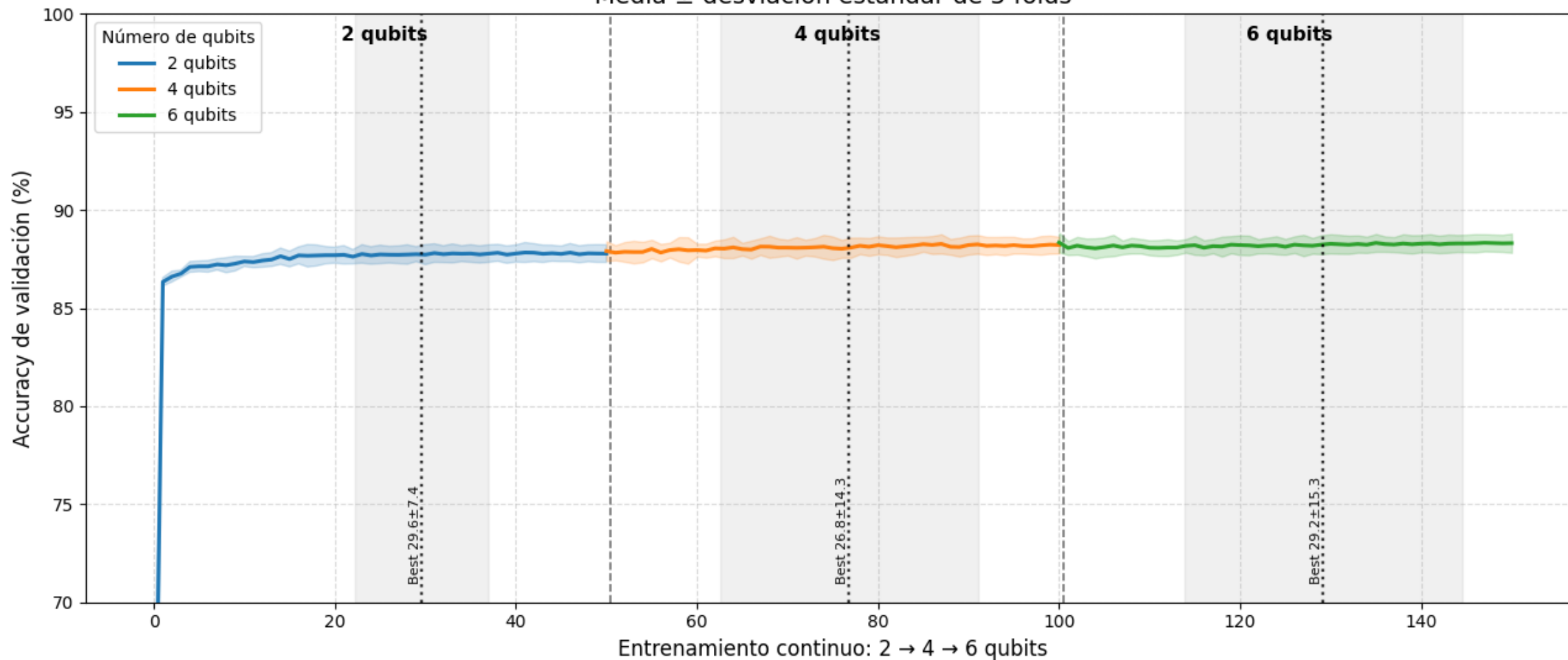
Loss de validación — AE no lineal
Media $\pm$ desviación estándar de 5 folds





AE no lineal

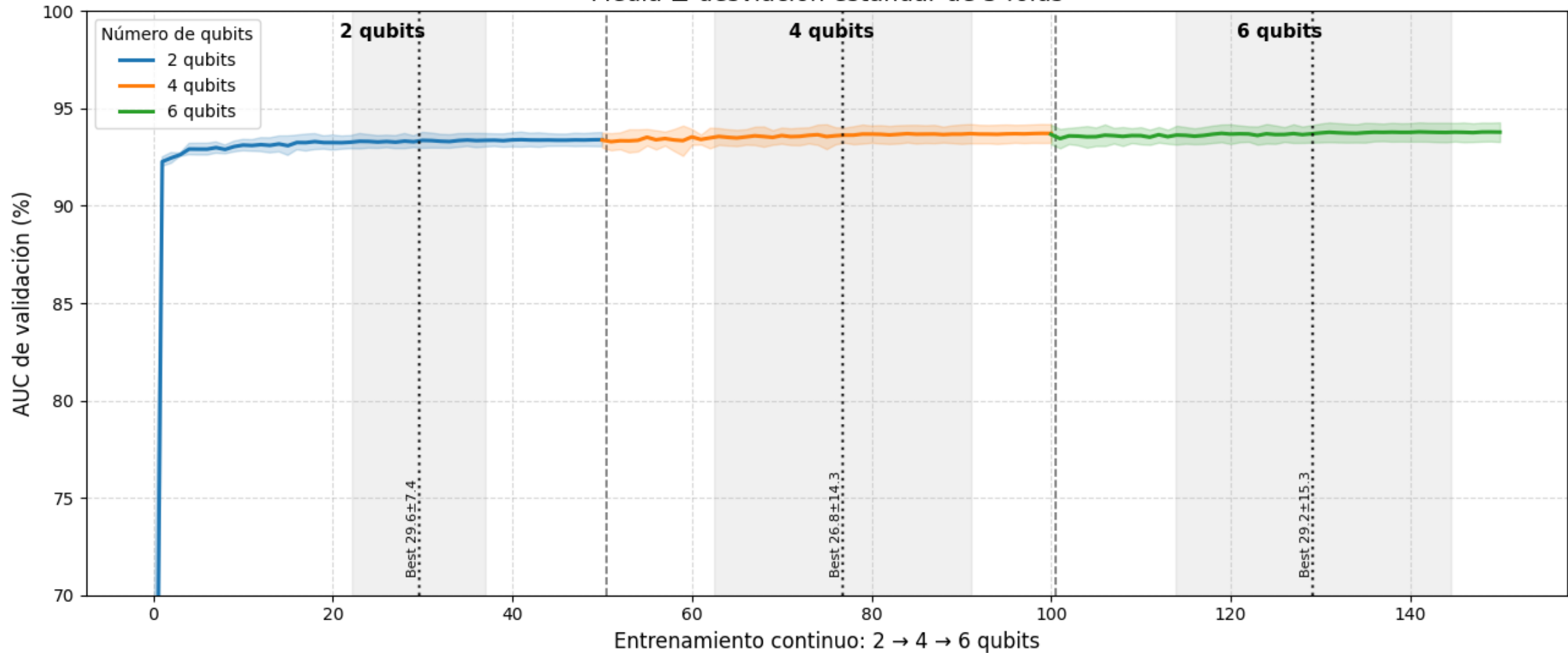
Accuracy de validación (%) — AE no lineal
Media $\pm$ desviación estándar de 5 folds





AE no lineal

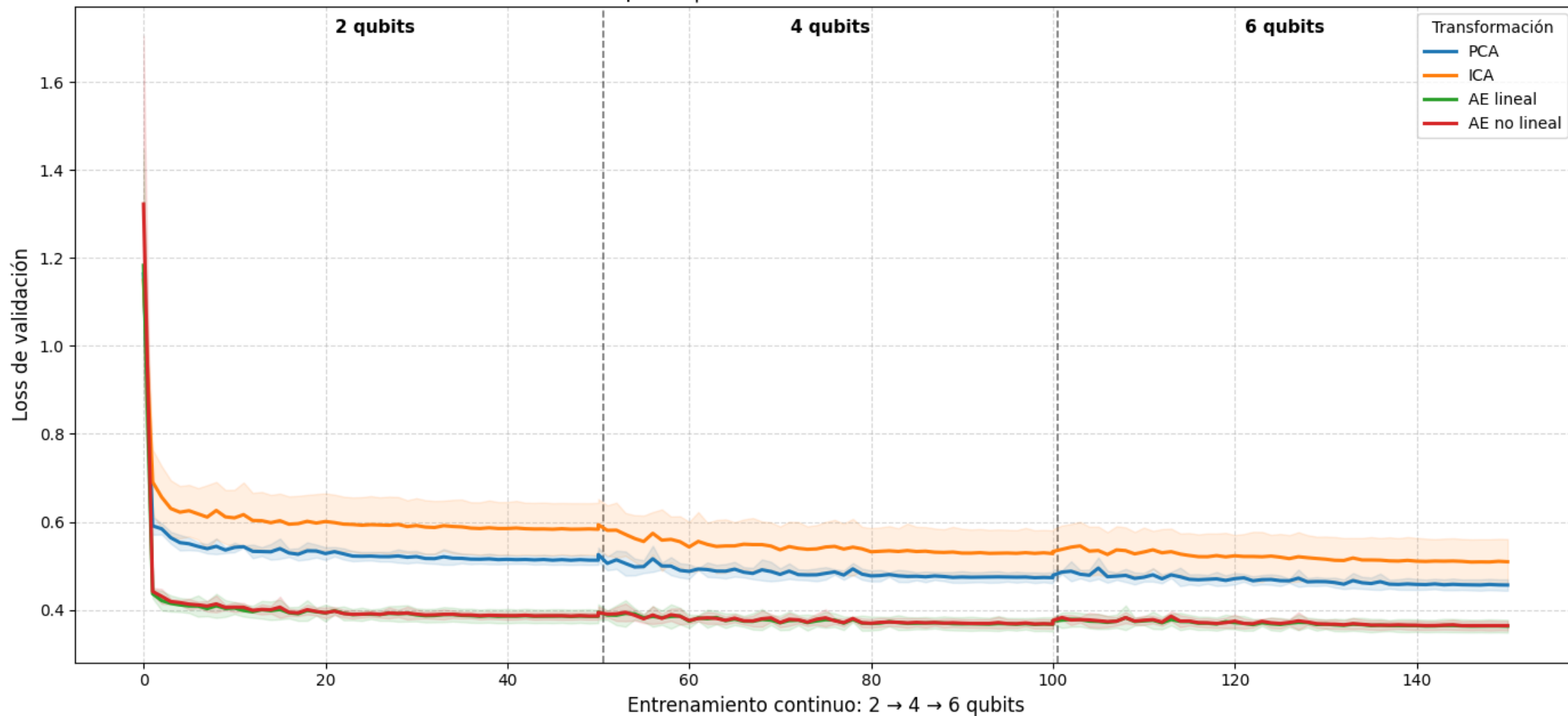
AUC de validación (%) — AE no lineal
Media $\pm$ desviación estándar de 5 folds





Graficas grupales

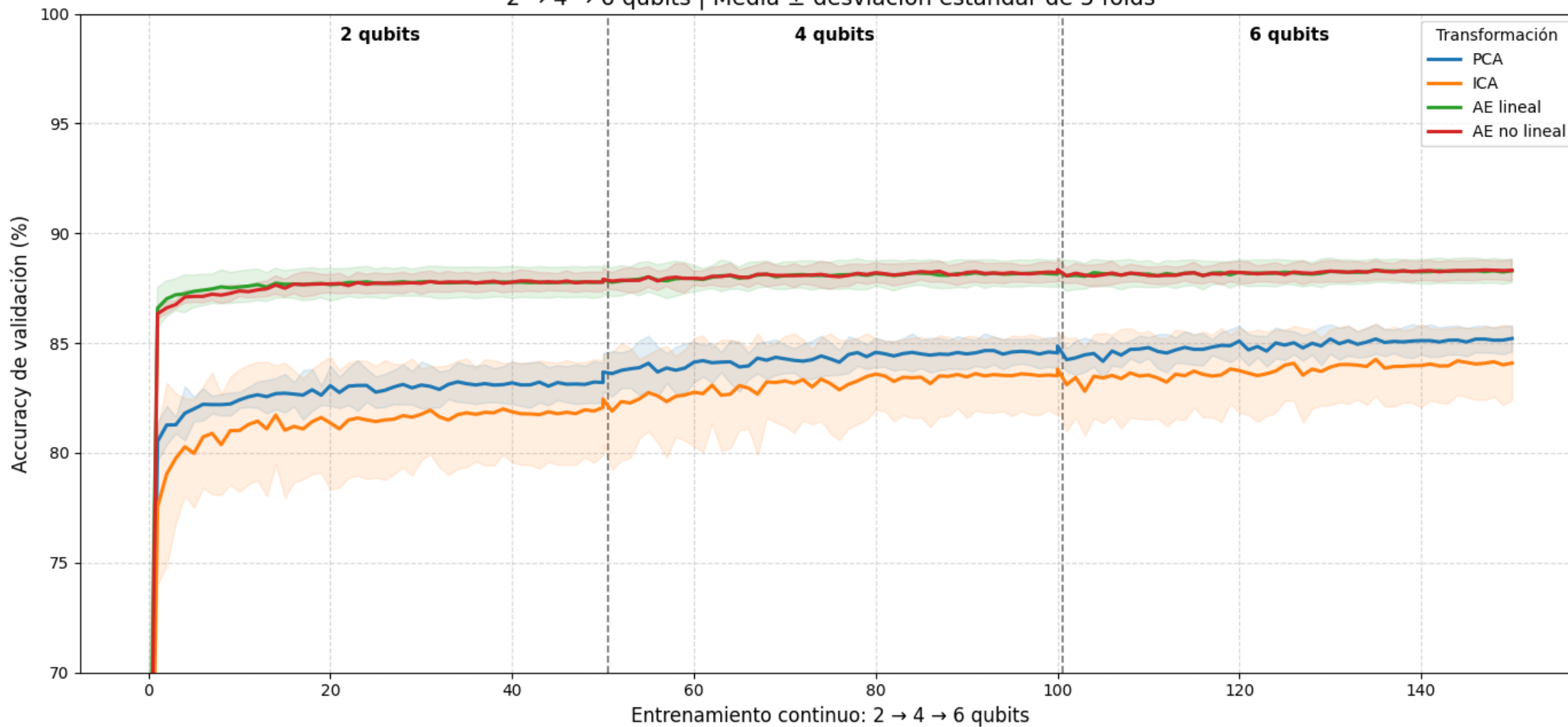
Loss de validación — Comparación general
2 → 4 → 6 qubits | Media $\pm$ desviación estándar de 5 folds





Graficas grupales

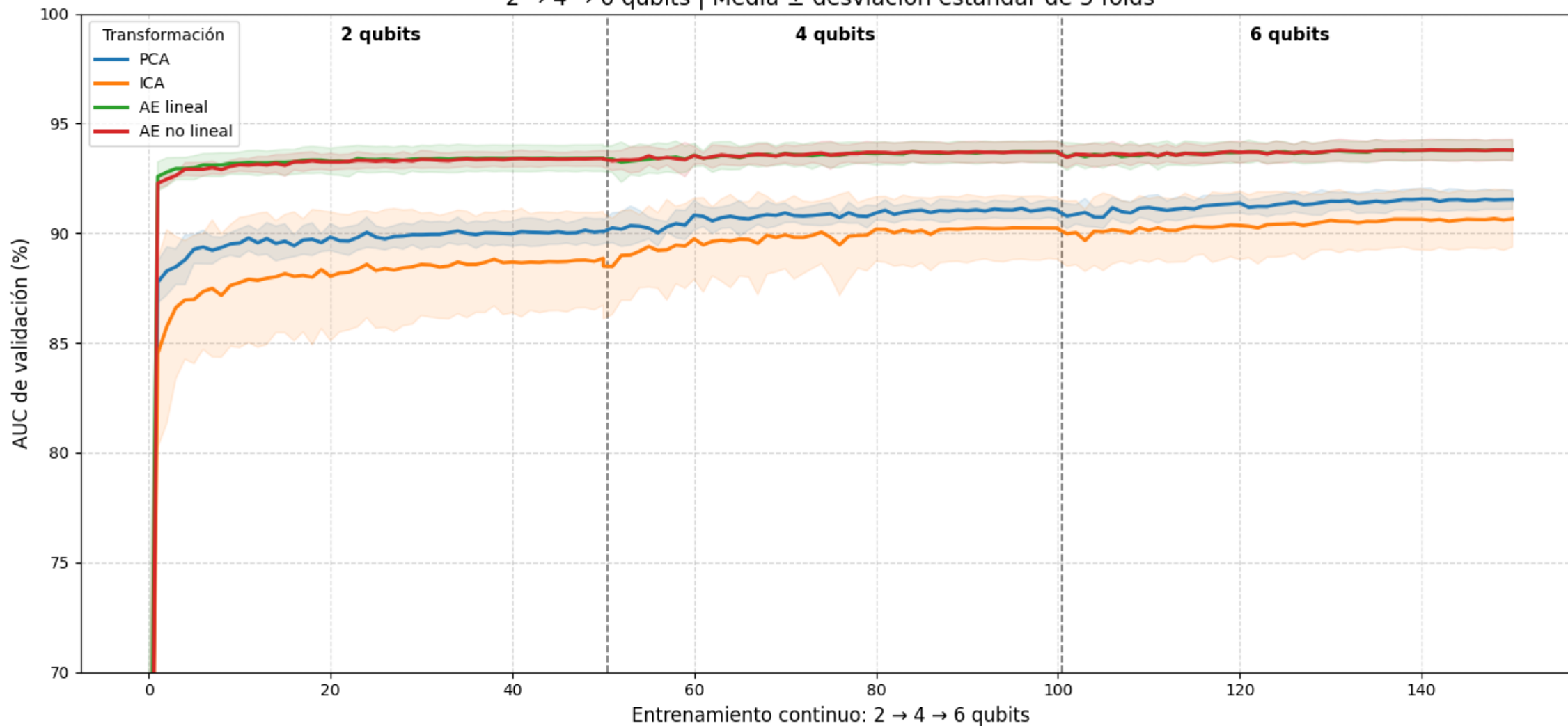
Accuracy de validación (%) — Comparación general
2 → 4 → 6 qubits | Media ± desviación estándar de 5 folds

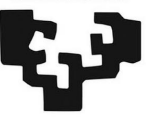




Graficas grupales

AUC de validación (%) — Comparación general
2 → 4 → 6 qubits | Media $\pm$ desviación estándar de 5 folds



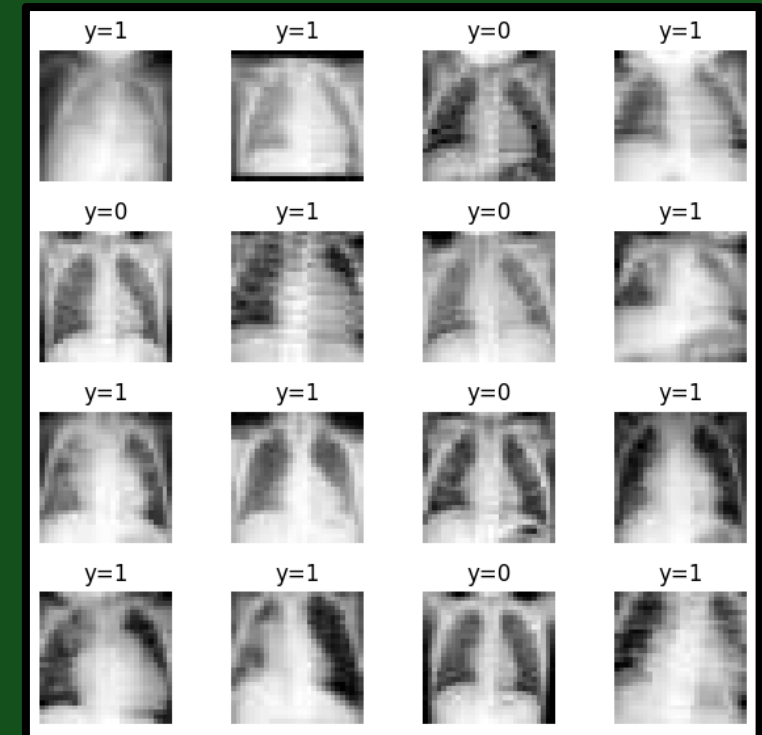


Contenido:

- Experimento 1.
- Experimento 2.
- Experimento 3.
- **Experimento 4.**

Modelos Híbridos

- Futuras pruebas.





HYBRID QUANTUM CONVOLUTIONAL NEURAL NETWORKS
MODEL FOR COVID-19 PREDICTION USING CHEST X-RAY
IMAGES

A PREPRINT

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February 15, 2021

ABSTRACT

Despite the great efforts to find an effective way for COVID-19 prediction, the virus nature and mutation represent a critical challenge to diagnose the covered cases. However, developing a model to predict COVID-19 via Chest X-Ray (CXr) images with accurate performance is necessary to help in early diagnosis. In this paper, a hybrid quantum-classical convolutional Neural Networks (HQCNN) model used the random quantum circuits (RQCs) as a base to detect COVID-19 patients with CXr images. A collection of 6952 CXr images, including 1161 COVID-19, 1575 normal, and 3216 pneumonia images, were used as a dataset in this work. The proposed HQCNN model achieved higher performance with an accuracy of 98.4% and a sensitivity of 99.3% on the first dataset cases. Besides, it obtained an accuracy of 99% and a sensitivity of 99.7% on the second dataset cases. Also, it achieved accuracy, and sensitivity of 88.6%, and 88.7%, respectively, on the third multi-class dataset cases. Furthermore, the HQCNN model outperforms various models in balanced accuracy, precision, F1-measure, and AUC-ROC score. The experimental results are achieved by the proposed model prove its ability in predicting positive COVID-19 cases.

Model	ACC.(%)	Sns.(%)	Spe.(%)	Prc.(%)	F1-measure(%)
HQCNN	98.4	99.3	97.8	96.7	98
CNN	96.8	98.6	95.7	93.7	96
MLP	92.9	100	88.4	84.8	91.7
CNN+SVM(RBF)	94.5	98.6	91.8	88.6	93.4
CNN+SVM(Poly)	93.5	99.3	89.7	86.2	92.3
CNN+KNN	95.5	98.6	93.5	90.8	94.6
CNN+AdaBoost	94.5	99.3	91.4	88.2	93.4
CNN+RF	94.2	98.6	91.4	88.1	93.1
CNN+BC	94.5	98.6	91.8	88.6	93.4
CNN+DT	95	99.3	92.3	89.2	94
CNN+GNB	94.8	98.6	92.3	89.2	93.7
CNN+XGBoost	95	99.3	92.3	89.2	94

Houssein, E. H., Abohashima, Z., Elhoseny, M., & Mohamed, W. M. (2022). Hybrid quantum-classical convolutional neural network model for COVID-19 prediction using chest X-ray images. *Journal of Computational Design and Engineering*, 9(2), 343–363.

Quantum-classical convolutional neural networks in
radiological image classification

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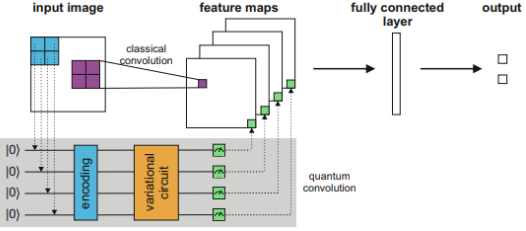
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Abstract—Quantum machine learning is receiving significant attention currently, but its usefulness in comparison to classical machine learning techniques for practical applications remains unclear. However, there are indications that certain quantum machine learning algorithms might result in improved training capabilities with respect to their classical counterparts - which might be particularly beneficial in situations with little training data available. Such situations naturally arise in medical classification tasks. Within this paper, different hybrid quantum-classical convolutional neural networks (QCCNN) with varying quantum circuit designs and encoding techniques are proposed. They are applied to two- and three-dimensional medical imaging data, e.g. featuring different, potentially malignant, lesions in computed tomography scans. The performance of these QCCNNs is already similar to the one of their classical counterparts - therefore encouraging further studies towards the direction of applying these algorithms within medical imaging tasks.

Index Terms—quantum computing, quantum machine learning, convolutional neural networks, imaging, medical classification, CT scans

Most of these studies, however, have been performed on artificial problem settings or on toy datasets, on which also empirical advantages of the QML variants over their classical counterparts were reported. Wide applications to real-world problems or to industrial use-cases are missing. More importantly, also a theoretical understanding of cases when quantum algorithms might show an advantage over classical algorithms is lacking. This situation partly arises from the lack of error-corrected quantum computers (QC) of large scale (i.e. with many error-corrected qubits), as well as from a missing routine to achieve a Quantum Random Access Memory (QRAM) to read in big amounts of (classical) data. Instead, currently available quantum computers are noisy and therefore referred to as *Noisy Intermediate Scale Quantum* (NISQ) devices [6]. These NISQ computers feature a limited number of noisy qubits of about 100, show a limited connectivity between the qubits



Matic, A., Monnet, M., Lorenz, J. M., Schachtner, B., & Messerer, T. (2022). *Quantum-classical convolutional neural networks in radiological image classification*. ArXiv.org. <https://arxiv.org/abs/2204.12390>

Motivación.

Quantum machine learning for image classification

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Image classification, a pivotal task in multiple industries, faces computational challenges due to the burgeoning volume of visual data. This research addresses these challenges by introducing two quantum machine learning models that leverage the principles of quantum mechanics for effective computations. Our first model, a hybrid quantum neural network with parallel quantum circuits, enables the execution of computations even in the noisy intermediate-scale quantum era, where circuits with a large number of qubits are currently infeasible. This model demonstrated a record-breaking classification accuracy of 99.21% on the full MNIST dataset, surpassing the performance of known quantum-classical models, while having eight times fewer parameters than its classical counterpart. Also, the results of testing this hybrid model on a Medical MNIST (classification accuracy over 99%), and on CIFAR-10 (classification accuracy over 82%), can serve as evidence of the generalizability of the model and highlights the efficiency of quantum layers in distinguishing common features of input data. Our second model introduces a hybrid quantum neural network with a Quantum convolutional layer, reducing image resolution via a convolution process. The model matches the performance of its classical counterpart, having four times fewer trainable parameters, and outperforms a classical model with equal weight parameters. These models represent advancements in quantum machine learning research and illuminate the path towards more accurate image classification systems.

Introduction

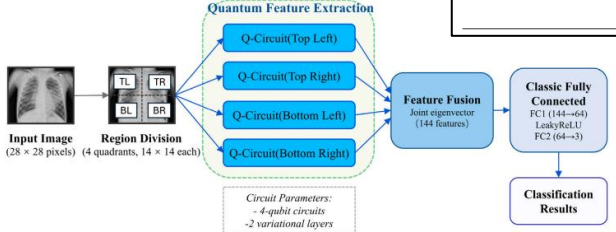
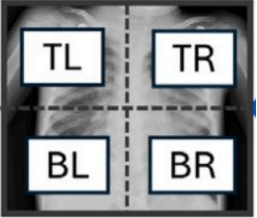
Image classification is a critical task in the modern world due to its wide range of practical applications in various fields [1]. For instance, in medical imaging, image classification algorithms have been shown to significantly improve the accuracy and speed of diagnoses of many diseases [2, 3]. In the field of autonomous vehicles, image classification plays a crucial role in object detection, tracking, and classification, which is necessary for safe and efficient navigation.

Deep learning approaches [4] like deep convolutional neural networks (CNNs) have emerged as powerful tools for image classification and recognition tasks [5, 6], achieving state-of-the-art performance on various benchmark datasets [7, 8]. However, as the amount of visual data grows exponentially, the computational cost of training

This potential arises from the unique features of quantum computing, such as superposition and entanglement, which can provide an exponential speedup for specific machine learning tasks [17]. QML algorithms produce probabilistic results, which align well with classification problems [18]. They also operate in an exponentially larger search space, which has the potential to enhance their performance [19–21]. However, it's important to note that the realization of these advantages in practical applications remains an active area of research and investigation. However, the real-world implementation of quantum algorithms faces significant challenges, such as the need for error correction and the high sensitivity of quantum systems to external disturbances [22]. Despite these challenges, QML has shown promising results in several applications [23]. In the context of image classification, QML algorithms can process large datasets of

Dataset	Model	train loss	test loss	test acc	param num
MNIST	CNN	0.0205	0.0449	98.71	372234
	HQNN	0.0204	0.0274	99.21	45194
Medical MNIST	CNN	$0.456 \cdot 10^{-2}$	$0.396 \cdot 10^{-2}$	82.64	247642
	HQNN	$0.429 \cdot 10^{-2}$	$0.332 \cdot 10^{-2}$	82.78	247462
CIFAR-10	CNN	0.3659	0.5484	82.78	81698
	HQNN	0.3851	0.5208	82.64	81578

Senokosov, A., Sedykh, A., Saginalieva, A., Kyriacou, B., & Melnikov, A. (2024). Quantum machine learning for image classification. *Machine Learning: Science and Technology*, 5(1), 015040.



A quantum-inspired medical scalable convolutional neural network for
Intelligent pneumonia diagnosis

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ARTICLE INFO

Keywords:
Quantum deep learning
Quantum computing
Artificial intelligence
Medical image processing
Convolutional neural network

ABSTRACT

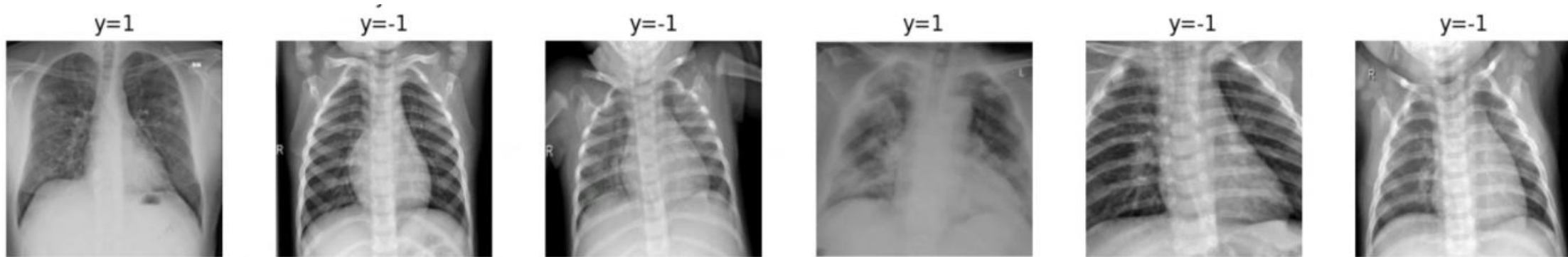
Biomedical images play a crucial role in clinical diagnosis, patient detection and management. Traditional medical image classification methods usually rely on manual feature extraction, which is difficult to capture complex image features. Moreover, some traditional algorithms are computationally inefficient and have long processing time when dealing with large-scale medical image data. Given the limitations of traditional methods, deep learning, especially convolutional neural networks (CNNs), has become a powerful tool for modern medical image classification. By automatically learning image features, convolutional neural networks can significantly improve the classification accuracy and efficiently process complex medical image data. Despite the excellent performance of convolutional neural networks in medical image classification, their computational resource requirements are still high. To address this problem, we propose a novel model, a quantum-inspired medical scalable convolutional neural network (MSQCNN). The MSQCNN system consists of a number of small-scale quantum devices to achieve full utilization of quantum resources and alleviate the present-day difficulty of preparing large quantum computers. We train this network on an image dataset containing chest radiographs and benchmark its performance. Compared with other benchmark models, our model performs even better. MSQCNN not only achieves an average accuracy of 97.71% on the test set, but also obtains an F1 score of 98.46%, an AUC value of 99.47%, a recall of 98.43%, and a precision of 98.50%. Moreover, we conclude that MSQCNN models have excellent resilience to noise by numerical simulations.

Li, D., Sun, Y., Yuan, Y., Hu, Z., Xiang, Q., Jiang, Y., Zhu, Y., Fu, Y., & Hua, X. (2026). A quantum-inspired medical scalable convolutional neural network for Intelligent pneumonia diagnosis. *Biomedical Signal Processing and Control*, 112, 108440.



Modelos Híbridos para imágenes

Notebook	Arquitectura	Extracción	Clasificador
01_CNN_MLP	CNN	CNN	MLP
02_CNN_z3_SU2	CNN	CNN ($z = 3$)	QNN SU(2)
03_CNN_z15_SU4_	CNN	CNN ($z = 15$)	QNN SU(4)
04_MSHQCNN_MLP	MS-HQCNN	MS-HQCNN	MLP
05_MSHQCNN_z3	MS-HQCNN	MS-HQCNN ($z = 3$)	QNN SU(2)
06_MSHQCNN_z15	MS-HQCNN	MS-HQCNN ($z = 15$)	QNN SU(4)





Rendimiento en test (umbral base)

Nº	Modelo	Loss	AUC	PR-AUC	Accuracy	F1	Balanced Accuracy
01	CNN → MLP	0.0332	0.9989	0.9977	98.68%	97.67%	98.61%
02	CNN → z3 → QNN SU(2)	0.0330	0.9990	0.9979	99.12%	98.44%	98.91%
03	CNN → z15 → QNN SU(4)	0.0273	0.9996	0.9991	99.34%	98.82%	99.07%
04	MS-HQCNN → MLP	0.0631	0.9981	0.9950	96.93%	94.31%	95.01%
05	MS-HQCNN → z3 → QNN SU(2)	0.0848	0.9986	0.9965	96.93%	94.31%	95.01%
06	MS-HQCNN → z15 → QNN SU(4)	0.0825	0.9981	0.9953	98.03%	96.41%	96.96%

Número de parámetros

Nº	Modelo	Parámetros totales	Parámetros clásicos	Parámetros cuánticos	Q. Extractor	Q. Clasificador
01	CNN → MLP	39,377	39,377	0	0	0
02	CNN → z3 → QNN SU(2)	35,141	35,123	18	0	18
03	CNN → z15 → QNN SU(4)	35,624	35,519	105	0	105
04	MS-HQCNN → MLP	6,297	6,249	48	48	0
05	MS-HQCNN → z3 → QNN SU(2)	976	910	66	48	18
06	MS-HQCNN → z15 → QNN SU(4)	1,471	1,318	153	48	105

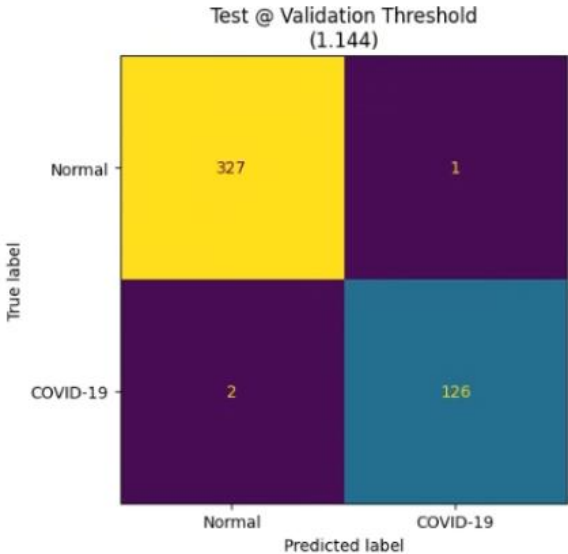
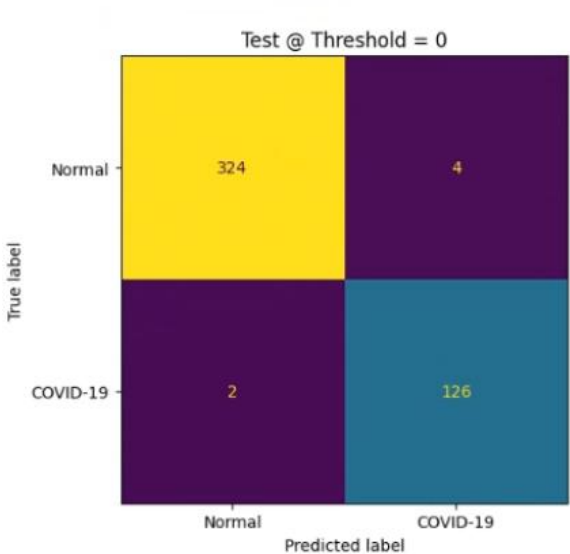
Familia	Parámetros totales	Reducción respecto a CNN → MLP
CNN → MLP	39,377	—
CNN → z3 → SU(2)	35,141	10.8 %
CNN → z15 → SU(4)	35,624	9.5 %
MS-HQCNN → MLP	6,297	84.0 %
MS-HQCNN → z3 → SU(2)	976	97.5 %
MS-HQCNN → z15 → SU(4)	1,471	96.3 %



01

=====				
TEST @ UMBRAL = 0				
=====				
	precision	recall	f1-score	support
Normal	0.99	0.99	0.99	328
COVID-19	0.97	0.98	0.98	128
accuracy			0.99	456
macro avg	0.98	0.99	0.98	456
weighted avg	0.99	0.99	0.99	456

=====				
TEST @ UMBRAL DE VALIDACIÓN (1.144005)				
=====				
	precision	recall	f1-score	support
Normal	0.99	1.00	1.00	328
COVID-19	0.99	0.98	0.99	128
accuracy			0.99	456
macro avg	0.99	0.99	0.99	456
weighted avg	0.99	0.99	0.99	456



Accuracy desde la matriz:
Threshold = 0 : 98.6842%
Threshold validación : 99.3421%



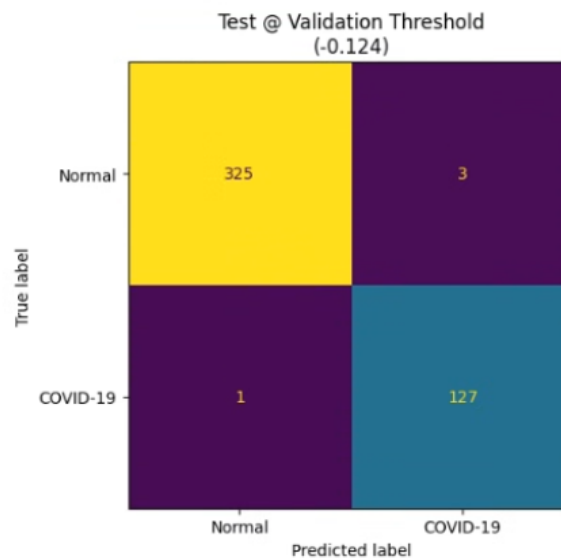
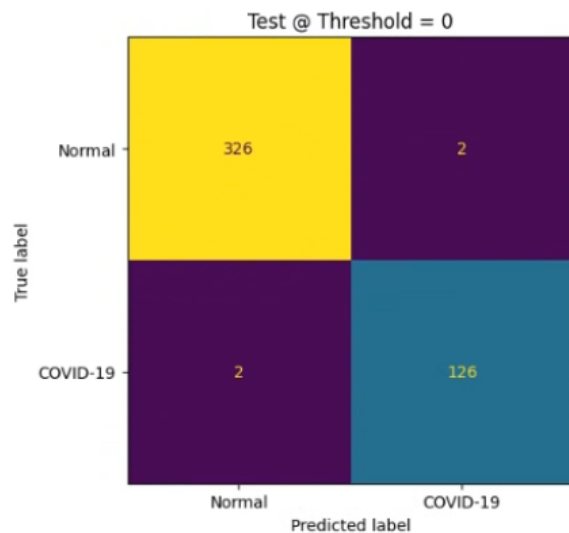
02

```
=====
TEST @ UMBRAL = 0
=====
```

	precision	recall	f1-score	support
Normal	0.99	0.99	0.99	328
COVID-19	0.98	0.98	0.98	128
accuracy			0.99	456
macro avg	0.99	0.99	0.99	456
weighted avg	0.99	0.99	0.99	456

```
=====
TEST @ UMBRAL DE VALIDACIÓN (-0.123562)
=====
```

	precision	recall	f1-score	support
Normal	1.00	0.99	0.99	328
COVID-19	0.98	0.99	0.98	128
accuracy			0.99	456
macro avg	0.99	0.99	0.99	456
weighted avg	0.99	0.99	0.99	456



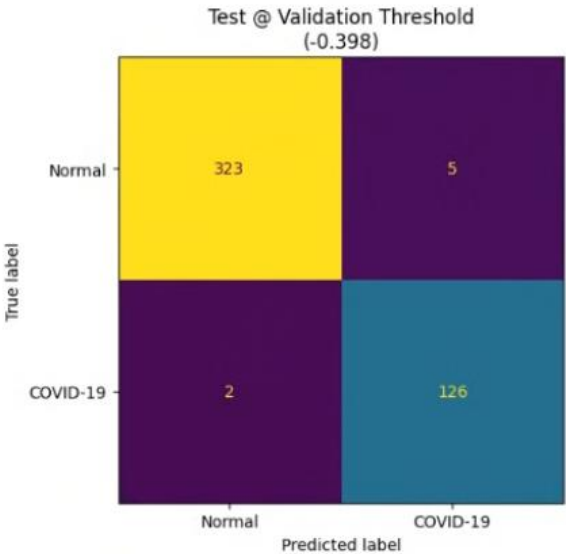
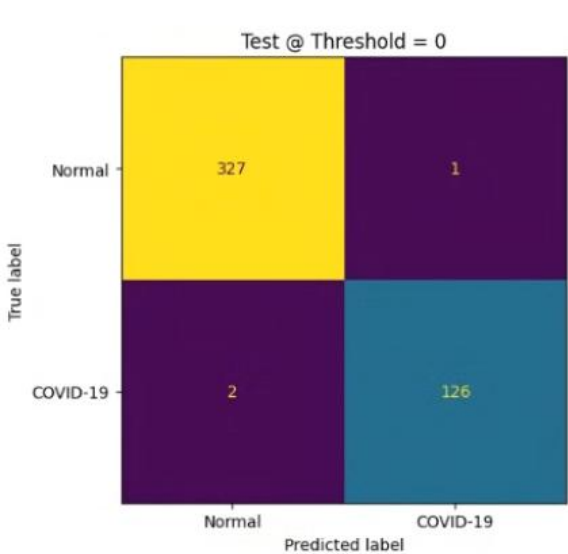
Accuracy desde la matriz:
Threshold = 0 : 99.1228%
Threshold validación : 99.1228%



03

=====				
TEST @ UMBRAL = 0				
=====				
	precision	recall	f1-score	support
Normal	0.99	1.00	1.00	328
COVID-19	0.99	0.98	0.99	128
accuracy			0.99	456
macro avg	0.99	0.99	0.99	456
weighted avg	0.99	0.99	0.99	456

=====				
TEST @ UMBRAL DE VALIDACIÓN (-0.397766)				
=====				
	precision	recall	f1-score	support
Normal	0.99	0.98	0.99	328
COVID-19	0.96	0.98	0.97	128
accuracy			0.98	456
macro avg	0.98	0.98	0.98	456
weighted avg	0.98	0.98	0.98	456



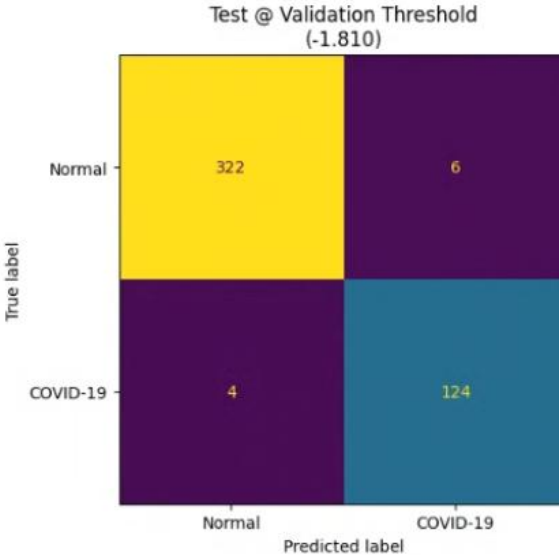
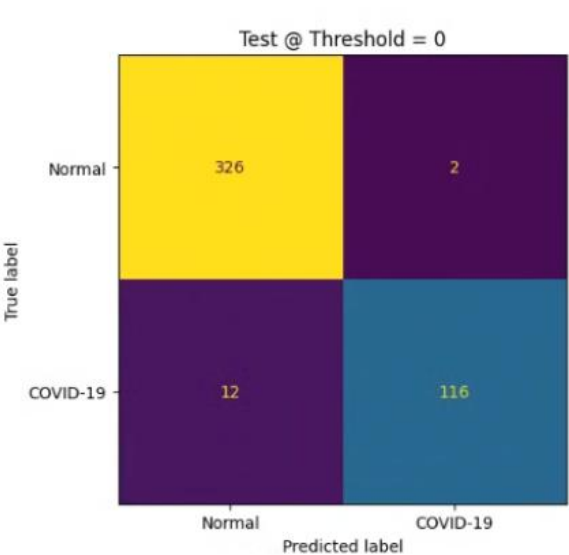
Accuracy desde la matriz:
Threshold = 0 : 99.3421%
Threshold validación : 98.4649%



04

TEST @ UMBRAL = 0				
	precision	recall	f1-score	support
Normal	0.96	0.99	0.98	328
COVID-19	0.98	0.91	0.94	128
accuracy			0.97	456
macro avg	0.97	0.95	0.96	456
weighted avg	0.97	0.97	0.97	456

TEST @ UMBRAL DE VALIDACIÓN (-1.809950)				
	precision	recall	f1-score	support
Normal	0.99	0.98	0.98	328
COVID-19	0.95	0.97	0.96	128
accuracy			0.98	456
macro avg	0.97	0.98	0.97	456
weighted avg	0.98	0.98	0.98	456



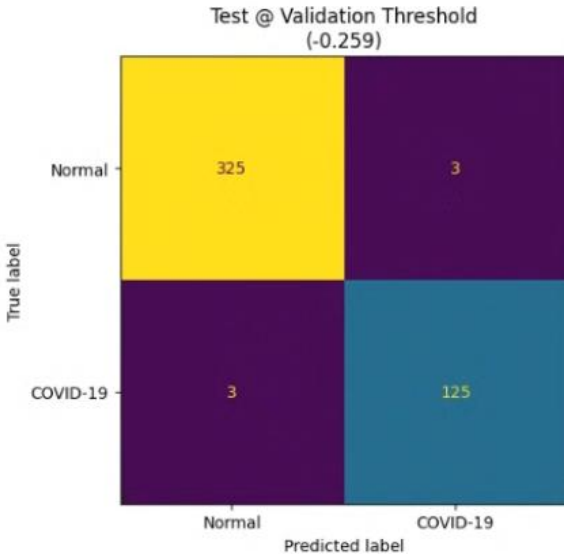
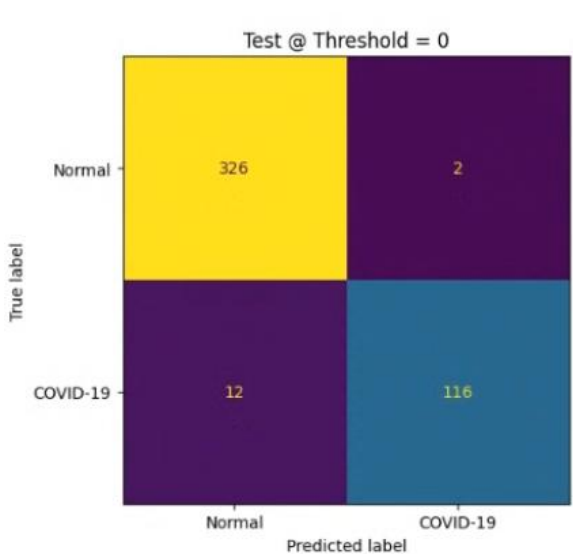
Accuracy desde la matriz:
Threshold = 0 : 96.9298%
Threshold validación : 97.8070%



05

=====				
TEST @ UMBRAL = 0				
=====				
	precision	recall	f1-score	support
Normal	0.96	0.99	0.98	328
COVID-19	0.98	0.91	0.94	128
accuracy			0.97	456
macro avg	0.97	0.95	0.96	456
weighted avg	0.97	0.97	0.97	456

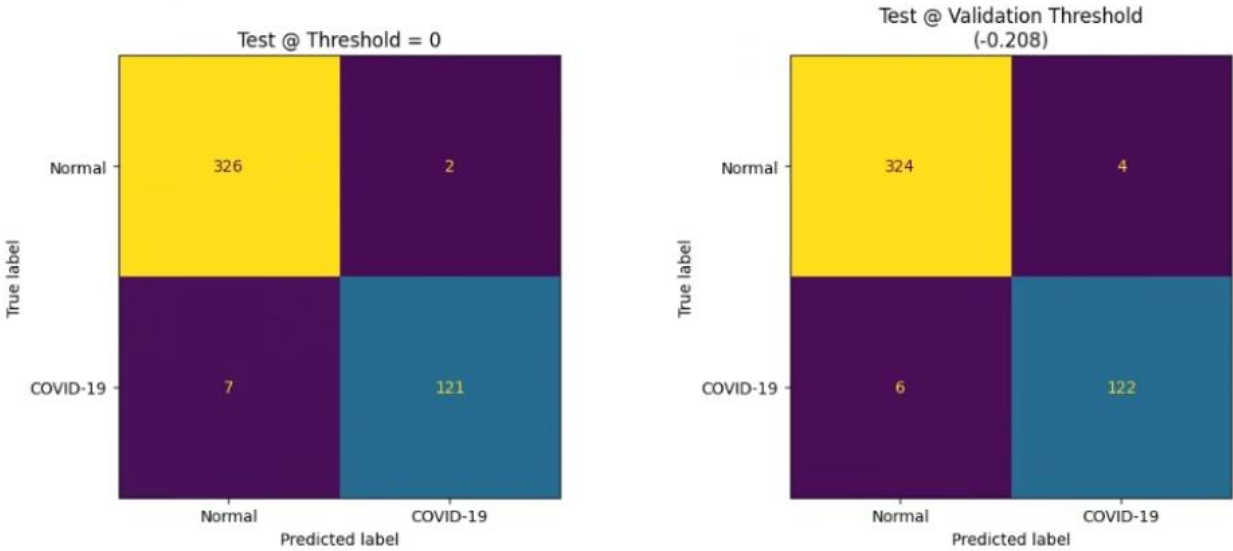
=====				
TEST @ UMBRAL DE VALIDACIÓN (-0.259363)				
=====				
	precision	recall	f1-score	support
Normal	0.99	0.99	0.99	328
COVID-19	0.98	0.98	0.98	128
accuracy			0.99	456
macro avg	0.98	0.98	0.98	456
weighted avg	0.99	0.99	0.99	456



Accuracy desde la matriz:
Threshold = 0 : 96.9298%
Threshold validación : 98.6842%

06

=====				
TEST @ UMBRAL = 0				
=====				
	precision	recall	f1-score	support
Normal	0.98	0.99	0.99	328
COVID-19	0.98	0.95	0.96	128
accuracy			0.98	456
macro avg	0.98	0.97	0.98	456
weighted avg	0.98	0.98	0.98	456
=====				
TEST @ UMBRAL DE VALIDACIÓN (-0.208410)				
=====				
	precision	recall	f1-score	support
Normal	0.98	0.99	0.98	328
COVID-19	0.97	0.95	0.96	128
accuracy			0.98	456
macro avg	0.98	0.97	0.97	456
weighted avg	0.98	0.98	0.98	456



Accuracy desde la matriz:
Threshold = 0 : 98.0263%
Threshold validación : 97.8070%

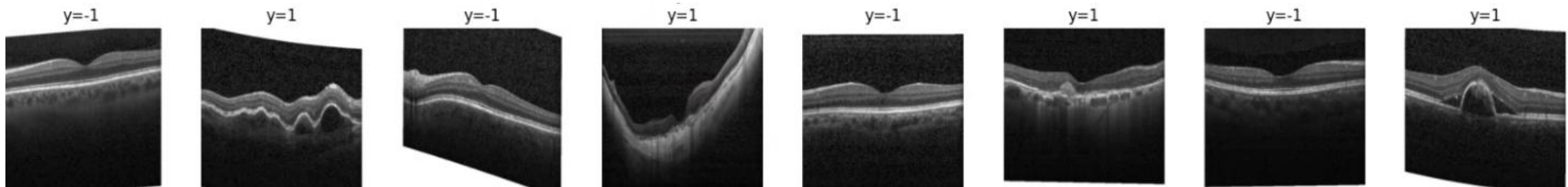


Contenido:

- Experimento 1.
- Experimento 2.
- Experimento 3.
- Experimento 4.
- Futuras pruebas.



- Modelos híbridos con NQK tanto $SU(2)$ como $SU(4)$
- Probar más conjuntos de imágenes.
- Probar más semillas para validar resultados.
- Explorar congelar las convoluciones.
- Implementar los bloques de $SU(4)$ a los modelos Híbridos.
- Comparar con la literatura.
- Identificar ventajas de cada modelo.





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